

FCEA v1.0

Federated Causal Evidence & Falsification Architecture

A domain-independent architecture for evidence-grounded causal analysis, falsification, scope control, reproducibility and auditable conclusions

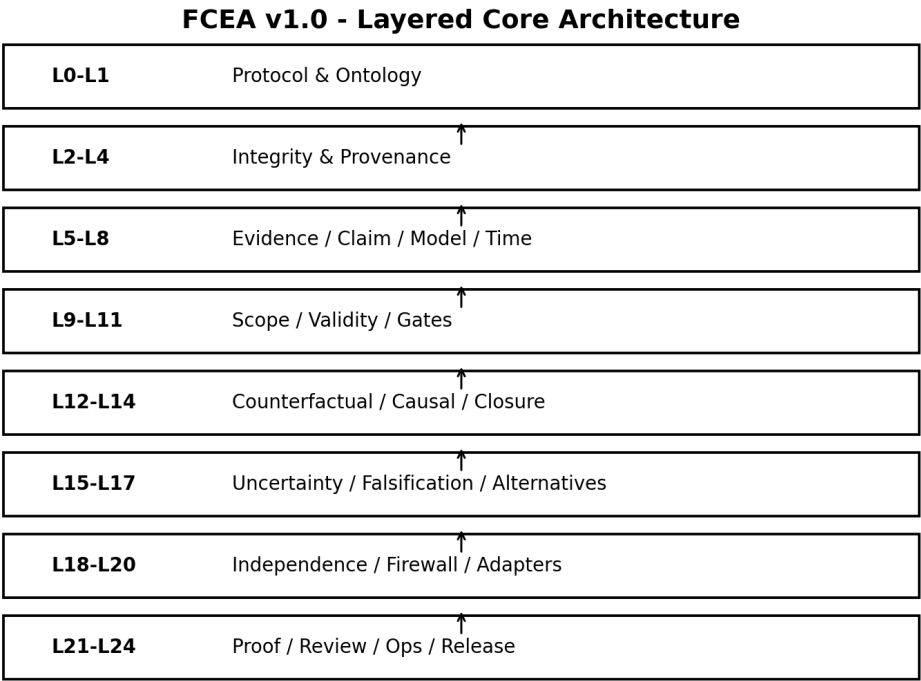


Figure 1. FCEA v1.0 high-level layer structure.

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Status: Complete Architecture and Implementation Standard

Scientific boundary: FCEA is a verification and falsification infrastructure. It does not transform incomplete evidence into certainty, and it does not replace domain expertise, legal process, scientific peer review, or empirical validation.

Executive Summary

FCEA v1.0 defines a reusable computational architecture for situations in which a material conclusion depends on many heterogeneous sources, models, assumptions, temporal relations and causal steps. Its purpose is to prevent a common failure mode in complex analysis: allowing a downstream conclusion to become stronger than the evidence and model chain that supports it.

The architecture treats evidence, claims, models, time, scope, counterfactuals, causal identification, closure, falsification, contradictory evidence, source independence, reproduction and human review as separate but interoperable objects. Every material output is required to carry an explicit domain, a validity state, an evidence state, a reproducibility record and a proof packet that permits a reviewer to traverse from the published statement back to the exact source objects and model runs.

FCEA is designed for scientific research, engineering assurance, public-policy evaluation, financial and market research, procurement and integrity analysis, climate attribution, public-health analysis, supply-chain resilience, geopolitical systems analysis and other domains where causal claims can be distorted by hidden assumptions, post-hoc thresholds, dependent sources, weak counterfactuals or uncontrolled scope widening.

What FCEA solves

- **Claim inflation:** a local, associational or model-specific result is prevented from being printed as a global or causal conclusion.
- **Provenance loss:** every material claim is linked to immutable source versions, transforms, assumptions and model runs.
- **Temporal leakage:** bitemporal knowledge control prevents future information from contaminating historical or prospective tests.
- **Counterfactual weakness:** causal claims must declare what would have happened under a relevant alternative and quantify uncertainty.
- **Confirmation bias:** high-impact hypotheses must face negative controls, placebo tests, competing explanations and contradictory evidence.
- **Source echo:** copied reporting is clustered so it is not miscounted as independent corroboration.
- **Reproducibility failure:** each model result is tied to a dataset release, code version, environment, random seed and run manifest.
- **Domain mismatch:** discipline-specific semantics are isolated in adapters without weakening the common evidence and scope invariants.

Primary users

User group	Primary benefit
Research teams	Preregistered hypotheses, transparent assumptions, reproducible model runs and explicit falsification routes.
Systems and safety architects	Layered validation gates, invariants, failure containment and auditable release decisions.
Auditors and oversight bodies	Traceable proof packets, immutable evidence lineage and explicit contradictory evidence.
Policy analysts	Counterfactual policy evaluation without automatically equating temporal association with causation.
Financial / market researchers	Holdout discipline, temporal leakage control, sensitivity analysis and model-risk separation.
Investigative analysts	Structured claim decomposition, source-dependency control and prohibition of unsupported intent or guilt inferences.
Scientific publishers / reviewers	Machine-readable evidence lineage, model assumptions, reproduction artifacts and scope-limited verdicts.
Public institutions	Transparent analytical processes that can preserve INCONCLUSIVE or UNRESOLVED outcomes instead of forcing decisions.

Normative governing principle

A system SHALL NOT emit a claim whose causal, evidentiary, temporal, model, or scope strength exceeds the weakest mandatory certificate required to support that claim.

This rule is formalized in FCEA as the Evidence Conservation Principle and enforced by the Gate Engine and Claim Escalation Firewall.

PART I - Mission, Scientific Contract and Formal Foundations

1. Mission and Design Objective

FCEA is designed as a neutral verification infrastructure rather than an accusation generator, theorem generator or confidence-score machine. The core receives a declared hypothesis, models, evidence and domain-specific adapter rules and returns the strongest conclusion that those objects jointly permit. A legitimate result may be SUPPORT, FALSIFICATION, INCONCLUSIVE, UNRESOLVED, OUTSIDE_DOMAIN or MODEL_INVALID. Preserving uncertainty is a required behavior, not a failure.

The system is particularly valuable where a result depends on a chain such as source -> interpretation -> mechanism -> causal effect -> downstream consequence. In such chains, one weak or missing mandatory link must cap the strength of the terminal conclusion.

1.1 Core research question

For any declared claim C and scope S, determine whether a complete, traceable and reproducible proof path exists from admissible evidence and validated models to C, while also testing explicit falsification conditions and competing explanations.

$\text{Decision}(C,S) = \text{Join}(\text{Validity}, \text{Evidence}, \text{Time}, \text{Scope}, \text{Identification}, \text{Falsification}, \text{Reproduction})$

The join operator is not an arithmetic average. It is a rule-based conjunction over mandatory dimensions, with blocking semantics for failed prerequisites.

1.2 Non-goals

- FCEA does not infer intent, wrongdoing or motive from association, profit, access, correlation or network centrality alone.
- FCEA does not certify a universal theory from one model, one parameter region, one experiment or one historical episode.
- FCEA does not treat model fit as proof of model truth.
- FCEA does not use a single generic probability of truth unless the domain adapter supplies a calibrated statistical interpretation and the required assumptions are satisfied.
- FCEA does not force a binary answer when evidence is incomplete or mutually contradictory.
- FCEA does not authorize use of unlawfully obtained data or circumvention of privacy, security or evidentiary law.

2. Scientific Status and Claim Discipline

FCEA distinguishes observation, evidence, model output, claim and verdict. This distinction is essential because a raw datum can be authentic while an interpretation derived from it is wrong; a model can be internally correct while applied outside its validity domain; and a statistically significant association can remain causally unidentified.

Object	Meaning	Cannot be silently treated as
Source	A raw document, dataset, measurement file, record or sensor stream.	Evidence sufficient for a conclusion.
EvidenceItem	A typed observation or extracted fact with provenance and uncertainty.	A causal conclusion.
Claim	A proposition with declared scope and required evidence.	A verified fact before gates pass.
Model	A mathematical, logical or computational representation.	Reality itself.

Verdict	A scoped result after gates, falsification and review.	A broader conclusion than its certificate.
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2.1 Epistemic states

State axis	Allowed states	Purpose
Execution	NOT_RUN, RUNNING, COMPLETE, BLOCKED, ERROR, NOT_IMPLEMENTED	Separates software/runtime condition from scientific outcome.
Scientific	FALSIFIED, NOT_SUPPORTED, SUPPORTED, CERTIFIED_WITHIN_SCOPE, INCONCLUSIVE, UNRESOLVED	Reports evidentiary state without implying software failure.
Validity	VALID, OUTSIDE_DOMAIN, MODEL_INVALID, DATA_INSUFFICIENT, PROVENANCE_FAILURE, TEMPORAL_FAILURE, IDENTIFICATION_FAILURE	Explains why a scientific result is or is not interpretable.

3. Formal System Definition

$$F = (H, D, E, C, M, T, S, G, K, X, V, P)$$

where H is the hypothesis set, D declared domains, E the Evidence Graph, C the Claim Graph, M the Model Graph, T the Temporal Graph, S the Scope Algebra, G gate records, K causal/counterfactual/closure operators, X falsification and contradiction tests, V verdict state, and P the Proof Packet.

A conforming implementation MUST preserve object identity and version history across these components. Derived objects may be superseded, but raw evidence and historical verdicts must remain recoverable for reproducibility and audit.

3.1 Evidence Conservation Principle

$$\text{Strength}(C) \leq \text{meet}_{\{d \text{ in } \text{Req}(C)\}} \text{Strength}(d)$$

The operator meet is the greatest lower bound in the relevant evidence-strength lattice. Because evidence strength is multidimensional, FCEA does not reduce it to one scalar by default.

$$Q(C) = (q_p, q_i, q_t, q_m, q_r, q_s)$$

where q_p = provenance quality, q_i = source independence, q_t = temporal validity, q_m = model validity, q_r = reproducibility and q_s = scope coverage. For mandatory dependencies D_i :

$$Q(C) \leq \text{meet}_i Q(D_i) \text{ (component-wise)}$$

This formulation prevents an apparently strong average from hiding a fatal weakness in one mandatory dimension.

4. Scope Algebra

Scope is not free text. It is a typed object representing where, when and under which assumptions a claim is permitted to apply.

$$S = (D, T, P, R, A, M, Q)$$

D is domain, T time interval, P population or system, R operating regime, A assumptions, M model family and Q precision/measurement regime. A partial order $S1 \leq S2$ means $S1$ is no broader than $S2$.

Claim($S1$) does not imply Claim($S2$) when $S1 < S2$ without a widening certificate.

Widening certificates may be supplied by exhaustive coverage, a valid theorem, external replication over the larger domain, or a statistically justified population inference, depending on adapter semantics.

4.1 Existential and universal asymmetry

Existential: exists x in D such that $P(x)$

Universal negative: for all x in D , not $P(x)$

A single valid witness may establish an existential statement, while repeated failure to find a witness does not establish the universal negative unless the search is exhaustive or a valid certificate covers the complete declared domain. This asymmetry is encoded directly in claim type and promotion rules.

PART II - Canonical Objects, Graphs and State Model

5. Canonical Object Model

Object	Minimum responsibility
SourceSpec	raw acquisition identity, origin, hash, legal basis, access route and immutable storage pointer
EvidenceItem	typed observation, value, uncertainty, source pointer, transform version and evidentiary role
HypothesisSpec	target proposition, null, alternatives, preregistered tests and falsification conditions
ClaimSpec	atomic proposition, scope, dependencies, allowed evidence and prohibited inferences
ModelSpec	equations/rules, parameters, assumptions, estimator, validity domain and provenance
ImplementationSpec	code package, environment, dependency lock, tests and mapping to ModelSpec
DomainSpec	space/time/population/system constraints and admissible operating regime
TemporalSpec	valid-time and knowledge-time semantics plus ordering constraints
ScopeSpec	typed scope tuple and widening rules
CounterfactualSpec	intervention/baseline definition, estimand, horizon, assumptions and uncertainty
CausalEstimandSpec	cause, outcome, population, intervention contrast and identification strategy
ClosureSpec	closure type, path semantics, orientation/order requirements and identity rule
TestSpec	method, threshold, expected output, negative/positive controls and preregistration metadata
GateRecord	gate input digest, outcome, blockers, reviewer and run metadata
VerdictRecord	scoped conclusion, states, evidence summary, prohibited claims and proof-packet reference
AdapterSpec	domain ontology, validity rules, time semantics, causal semantics, closure semantics and test suite
RunManifest	dataset release, code commit, container digest, parameters, seed, environment and hashes
ProofPacket	self-contained review and reproduction package for a material verdict

5.1 Example machine-readable ClaimSpec

```
{
  "claim_id": "CLM-2026-0042",
  "proposition": "Exposure X causally changes outcome Y",
  "claim_type": "CAUSAL_EFFECT",
  "scope_ref": "SCP-17",
  "required_dependencies": ["OBS-X", "OBS-Y", "MODEL-3", "CF-2", "ID-7"],
  "required_gates": ["G2", "G5", "G6", "G7", "G8", "G9", "G12", "G15"],
  "prohibited_inferences": ["intent", "universal_generalization"],
  "status": "UNRESOLVED"
}
```

6. Four Linked Graphs

Four Linked Graphs with Independent Scope and Gate Control

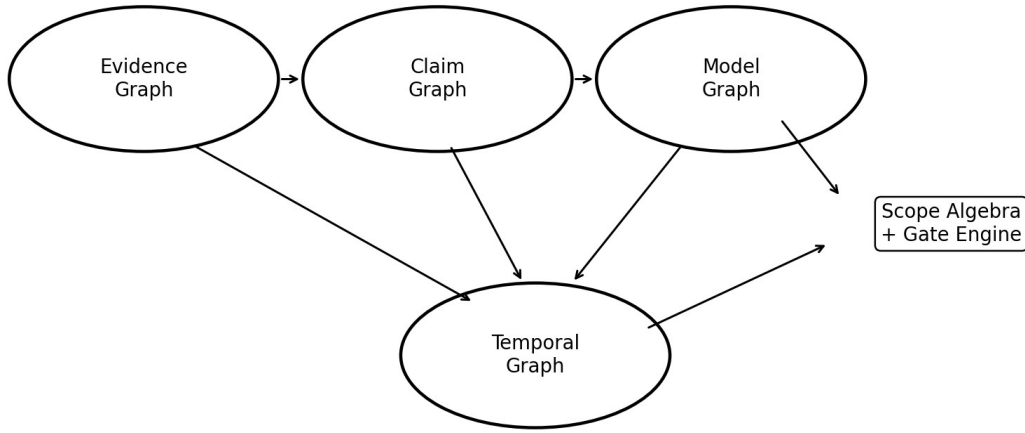


Figure 2. Evidence, Claim, Model and Temporal graphs remain distinct and are joined under Scope Algebra and Gate Engine control.

6.1 Evidence Graph

$$G_E = (V_E, E_E)$$

Nodes represent sources, observations, measurements, datasets, experiment outputs or derived evidence items. Edges are typed as supports, contradicts, neutral_to, derived_from, depends_on, duplicates or supersedes. Evidence-role labels are claim-relative: the same evidence may support one claim and contradict another.

6.2 Claim Graph

$$G_C = (V_C, E_C)$$

Complex conclusions are decomposed into atomic claims connected by requires, implies_candidate, refines, conflicts_with and generalizes edges. The graph makes missing proof steps visible. A terminal claim cannot be promoted if any mandatory dependency remains unresolved.

6.3 Model Graph

$$G_M = (V_M, E_M)$$

The Model Graph records mathematical models, alternative specifications, assumptions, estimands, implementations and validation tests. It supports derived_from, approximates, extends, competes_with, implements, calibrated_on and tested_on edges. This separation prevents a software implementation from being mistaken for the underlying scientific model.

6.4 Temporal Graph

$$G_T = (V_T, E_T)$$

The Temporal Graph represents events and ordering relations. Every relevant object may carry valid time t_v and knowledge time t_k . Historical reconstruction at cutoff t_c is restricted to objects satisfying $t_k \leq t_c$. The graph can represent precedes, overlaps, contains, contemporaneous_with and uncertain_order.

7. Bitemporal Integrity

$$\text{admissible}(o, t_c) = 1 \text{ if } t_k(o) \leq t_c, \text{ else } 0$$

Valid time describes when the underlying fact or state was true. Knowledge time describes when the analytical system could legitimately know it. They must never be collapsed. A backtest that uses a source published after the historical cutoff is invalid even if the source describes an earlier event.

8. Claim Dependency Semantics

A claim may have mandatory and optional dependencies. Mandatory dependencies are conjunctive by default. Optional evidence may improve robustness but cannot compensate for a missing mandatory certificate unless the adapter explicitly defines a logically valid alternative proof route.

$$\text{Req}(C) = \{D1, D2, \dots, Dn\}$$

$$\text{Eligible}(C) = \text{AND}_i \text{MandatoryPass}(D_i)$$

Multiple proof routes are represented as disjunctive normal form:

$$\text{Eligible}(C) = \text{OR}_r [\text{AND}_{\{d \in \text{Route}_r(C)\}} \text{Pass}(d)]$$

This representation avoids the mistake of averaging evidence from incompatible routes.

PART III - Core Architecture L0-L24

L0 - Protocol Freeze and System Boundary

Fix the analytical question, hypotheses, null, alternatives, observation window, cutoff, data universe, primary estimands, thresholds, falsification criteria and amendment policy before confirmatory analysis.

Mandatory controls

- Every confirmatory run references a frozen protocol version.
- Protocol amendments are append-only, justified and trigger a new analysis generation.
- Exploratory and confirmatory analyses are flagged separately.
- The system boundary lists explicitly excluded actors, variables, domains and data sources.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

Required outputs

Outputs SHALL be machine-readable and human-readable. At minimum they include a result state, scope, method/version, run identifier, evidence references, contradictory evidence if applicable, uncertainty or blocker status, and the exact downstream components permitted to consume the result.

Failure modes and safeguards

Typical failure modes include schema drift, hidden manual edits, post-cutoff leakage, model misuse, confirmation bias, source echo, silent scope widening and irreproducible computation. FCEA treats these as release-blocking when they affect a mandatory dependency. The default corrective action is downgrade, rerun or INCONCLUSIVE rather than forced promotion.

L1 - Typed Ontology and Semantic Contracts

Define canonical entity, relation, claim, evidence and model types so that semantically different objects cannot be combined by convenience.

Mandatory controls

- Units and dimensions are machine-checkable.
- Relationship predicates have declared directionality and admissible endpoint types.
- Association, ownership, influence, control and causation remain distinct relations.
- Adapters may extend but not redefine core meanings inconsistently.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

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L2 - Source Integrity and Immutable Acquisition

Acquire sources into an immutable raw store with cryptographic identifiers and acquisition metadata.

Mandatory controls

- Compute SHA-256 or stronger approved digest at acquisition.
- Preserve original bytes and source-native identifiers.
- Record acquisition timestamp, access route, legal basis and parser status.
- Any transformed representation receives a new identifier and retains its parent pointer.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

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```
source_digest = SHA256(raw_bytes)
```

L3 - Provenance and Chain-of-Custody Engine

Maintain reversible lineage from material verdicts to raw evidence and transformations.

Mandatory controls

- Every derived field names source fields and transform function/version.
- Manual annotations are first-class signed events, not silent edits.
- Deletion is logical where legally required; historical lineage records remain consistent with the applicable retention policy.
- A missing provenance path blocks material publication.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

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L4 - Normalization, Resolution and Data Quality

Normalize units and schemas while preserving raw values; resolve identities and quantify ambiguity.

Mandatory controls

- Never overwrite raw units or original spelling/identifiers.
- Entity resolution decisions are reversible and uncertainty-aware.
- Duplicate and near-duplicate evidence detection precedes corroboration counting.
- Quality checks distinguish data-quality anomalies from substantive anomalies.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

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L5 - Evidence Graph

Represent supporting, contradicting, neutral and dependent evidence as graph objects.

Mandatory controls

- Evidence role is claim-specific.
- Source dependency clusters prevent false independence.
- Evidence items carry uncertainty and observation status.
- High-impact claims require sufficient independent evidence clusters according to adapter policy.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

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L6 - Claim Graph

Decompose broad narratives into atomic, testable propositions and explicit dependency chains.

Mandatory controls

- Every material conclusion has a unique ClaimSpec.
- Claims declare required evidence, model and gate dependencies.
- The graph stores contradictions and alternative hypotheses.
- No claim becomes a finding without an auditable proof path.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

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Outputs SHALL be machine-readable and human-readable. At minimum they include a result state, scope, method/version, run identifier, evidence references, contradictory evidence if applicable, uncertainty or blocker status, and the exact downstream components permitted to consume the result.

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L7 - Model Graph

Represent theoretical, statistical, causal, numerical and mechanistic models separately from their implementations.

Mandatory controls

- Each ModelSpec declares assumptions, parameters and validity domain.
- ImplementationSpec must map code to a versioned model.
- Competing specifications are retained, not discarded after selection.
- Model-selection decisions are part of the run record.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

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L8 - Temporal Graph and Knowledge Cutoff

Enforce temporal ordering, interval semantics and knowledge-time constraints.

Mandatory controls

- Causal routes require adapter-defined cause-before-effect constraints.

- Future information cannot enter historical or prospective holdouts.
- Uncertain dates are represented as intervals/distributions, not arbitrary point dates.
- Contradictory timelines are explicit blockers where material.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

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$$\text{admissible}(o, t_c) = 1[t_k(o) \leq t_c]$$

L9 - Scope Algebra

Attach every claim, test and verdict to a typed scope and prevent unsupported widening.

Mandatory controls

- Outputs cannot have broader scope than their certificates.
- Generalization requires a named widening certificate.
- Existential and universal claim types use different promotion rules.
- Scope includes assumptions and model regime, not only population and dates.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

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$$\text{Scope}(\text{output}) \leq \text{Scope}(\text{certificate}), \text{ unless } \text{WideningCertificate} = \text{VALID}$$

L10 - Validity and Assumption Engine

Evaluate whether data, models and assumptions are valid for the declared domain before substantive interpretation.

Mandatory controls

- Validity predicates are executable where possible.
- Assumptions are not hidden in narrative prose only.
- Failed mandatory validity predicates produce explicit invalid/outside-domain states.
- Approximation and truncation errors are part of validity when applicable.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

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L11 - Gate Engine

Execute prerequisite gates and block downstream claims when required certificates fail or remain unresolved.

Mandatory controls

- Gate outcomes are PASS, FAIL, INCONCLUSIVE, BLOCKED or NOT_IMPLEMENTED.
- A failed gate does not silently disappear in aggregate scoring.
- Each gate stores inputs, digest, method, output and blocker reason.
- Gate dependencies form an acyclic control graph even when domain models contain feedback cycles.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

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$$\text{Advance}(C) = \text{AND}_{\{g \text{ in RequiredGates}(C)\}} [g = \text{PASS}]$$

L12 - Counterfactual Engine

Define relevant alternative worlds or baselines and estimate how conclusions change under them.

Mandatory controls

- Counterfactuals are versioned objects.
- One convenient baseline is not treated as the only plausible world.
- Counterfactual uncertainty propagates to causal conclusions.
- Physical/mechanistic adapters may use perturbation or alternative-model semantics rather than social-science do-operator semantics.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

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Outputs SHALL be machine-readable and human-readable. At minimum they include a result state, scope, method/version, run identifier, evidence references, contradictory evidence if applicable, uncertainty or blocker status, and the exact downstream components permitted to consume the result.

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$$ATE = E[Y(1) - Y(0)]$$

L13 - Causal Identification Engine

Require a stated estimand and identification strategy for causal claims.

Mandatory controls

- DAG or mechanistic dependency structure is explicit.
- Confounders, mediators, colliders and selection mechanisms are documented where applicable.
- Identification failure downgrades a causal claim to association or unresolved.
- Diagnostics and sensitivity tests are tied to the estimand, not generic model fit.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

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Outputs SHALL be machine-readable and human-readable. At minimum they include a result state, scope, method/version, run identifier, evidence references, contradictory evidence if applicable, uncertainty or blocker status, and the exact downstream components permitted to consume the result.

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$$\tau(x) = E[Y \mid \text{do}(X=x_1)] - E[Y \mid \text{do}(X=x_0)] \quad (\text{when SCM semantics are valid})$$

L14 - Closure Engine

Verify global path closure, feedback mechanisms or domain-specific closed structures without conflating different closure semantics.

Mandatory controls

- ClosureType is explicit.
- Every edge in a claimed closed mechanism must independently pass its required gates.
- Temporal orientation and identity rules are adapter-specific and mandatory.
- Graph-cycle detection alone is never sufficient to certify causal feedback.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

Required outputs

Outputs SHALL be machine-readable and human-readable. At minimum they include a result state, scope, method/version, run identifier, evidence references, contradictory evidence if applicable, uncertainty or blocker status, and the exact downstream components permitted to consume the result.

Failure modes and safeguards

Typical failure modes include schema drift, hidden manual edits, post-cutoff leakage, model misuse, confirmation bias, source echo, silent scope widening and irreproducible computation. FCEA treats these as release-blocking when they affect a mandatory dependency. The default corrective action is downgrade, rerun or INCONCLUSIVE rather than forced promotion.

$$\text{Closure}(\pi) = \text{Identity}(v_0, v_k) \text{ AND } \text{AND}_{\{e \text{ in } \pi\}} \text{Valid}(e) \text{ AND } \text{TemporalConstraints}(\pi)$$

L15 - Uncertainty Engine

Represent and propagate data, measurement, parameter, model, counterfactual, structural and scope uncertainty.

Mandatory controls

- Uncertainty components remain distinguishable.
- Monte Carlo or bootstrap results store seeds and sampling design.
- No uncalibrated single truth probability is emitted.
- Sensitivity surfaces are retained, not only the best-case point estimate.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

Required outputs

Outputs SHALL be machine-readable and human-readable. At minimum they include a result state, scope, method/version, run identifier, evidence references, contradictory evidence if applicable, uncertainty or blocker status, and the exact downstream components permitted to consume the result.

Failure modes and safeguards

Typical failure modes include schema drift, hidden manual edits, post-cutoff leakage, model misuse, confirmation bias, source echo, silent scope widening and irreproducible computation. FCEA treats these as release-blocking when they affect a mandatory dependency. The default corrective action is downgrade, rerun or INCONCLUSIVE rather than forced promotion.

$$U = (U_{\text{data}}, U_{\text{measurement}}, U_{\text{parameter}}, U_{\text{model}}, U_{\text{counterfactual}}, U_{\text{structural}}, U_{\text{scope}})$$

L16 - Falsification Engine

Attempt to disprove or weaken hypotheses through negative controls, placebos, adversarial tests and competing predictions.

Mandatory controls

- Each high-impact hypothesis declares incompatible observations.
- Tests that cannot weaken a hypothesis are classified as supportive diagnostics, not falsification tests.
- Multiple-testing control is mandatory when many hypotheses or windows are searched.
- Failed falsification tests may support a claim but do not automatically verify universal truth.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

Required outputs

Outputs SHALL be machine-readable and human-readable. At minimum they include a result state, scope, method/version, run identifier, evidence references, contradictory evidence if applicable, uncertainty or blocker status, and the exact downstream components permitted to consume the result.

Failure modes and safeguards

Typical failure modes include schema drift, hidden manual edits, post-cutoff leakage, model misuse, confirmation bias, source echo, silent scope widening and irreproducible computation. FCEA treats these as release-blocking when they affect a mandatory dependency. The default corrective action is downgrade, rerun or INCONCLUSIVE rather than forced promotion.

L17 - Contradiction and Alternative-Explanation Engine

Collect and evaluate evidence and models that could explain the observations without the target hypothesis.

Mandatory controls

- Contradictory evidence is a first-class object.
- Benign or null explanations must be testable where possible.
- Alternative models use the same frozen data cutoff.
- Unresolved major contradictions block strong promotion.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

Required outputs

Outputs SHALL be machine-readable and human-readable. At minimum they include a result state, scope, method/version, run identifier, evidence references, contradictory evidence if applicable, uncertainty or blocker status, and the exact downstream components permitted to consume the result.

Failure modes and safeguards

Typical failure modes include schema drift, hidden manual edits, post-cutoff leakage, model misuse, confirmation bias, source echo, silent scope widening and irreproducible computation. FCEA treats these as release-blocking when they affect a mandatory dependency. The default corrective action is downgrade, rerun or INCONCLUSIVE rather than forced promotion.

L18 - Evidence Independence Engine

Estimate effective corroboration after accounting for shared upstream sources and duplicated reporting.

Mandatory controls

- Build a Source Dependency Graph.
- Cluster evidence derived from a common primary source.
- Nominal source count is never equated to independent source count.
- Independence assumptions are exposed in the Proof Packet.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

Required outputs

Outputs SHALL be machine-readable and human-readable. At minimum they include a result state, scope, method/version, run identifier, evidence references, contradictory evidence if applicable, uncertainty or blocker status, and the exact downstream components permitted to consume the result.

Failure modes and safeguards

Typical failure modes include schema drift, hidden manual edits, post-cutoff leakage, model misuse, confirmation bias, source echo, silent scope widening and irreproducible computation. FCEA treats these as release-blocking when they affect a mandatory dependency. The default corrective action is downgrade, rerun or INCONCLUSIVE rather than forced promotion.

$$N_{\text{effective}} \leq N_{\text{nominal}}$$

L19 - Claim Escalation Firewall

Prevent prohibited inferential jumps and enforce the Evidence Conservation Principle.

Mandatory controls

- Correlation does not become causation without identification.
- Benefit does not become intent without dedicated evidence.
- Access does not become influence, and influence does not become behavior automatically.
- Local evidence does not become global theory-level proof.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

Required outputs

Outputs SHALL be machine-readable and human-readable. At minimum they include a result state, scope, method/version, run identifier, evidence references, contradictory evidence if applicable, uncertainty or blocker status, and the exact downstream components permitted to consume the result.

Failure modes and safeguards

Typical failure modes include schema drift, hidden manual edits, post-cutoff leakage, model misuse, confirmation bias, source echo, silent scope widening and irreproducible computation. FCEA treats these as release-blocking when they affect a mandatory dependency. The default corrective action is downgrade, rerun or INCONCLUSIVE rather than forced promotion.

$$\text{Strength}(\text{output}) \leq \text{meet Strength}(\text{mandatory_dependencies})$$

L20 - Federated Adapter Framework

Isolate domain-specific ontology, validity, causal and closure semantics while preserving core invariants.

Mandatory controls

- Adapters may add stricter gates.
- Adapters may not disable provenance, scope control, contradictory evidence or reproducibility.
- Adapters provide their own test suite and conformance manifest.
- A nonconforming adapter is rejected before production use.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

Required outputs

Outputs SHALL be machine-readable and human-readable. At minimum they include a result state, scope, method/version, run identifier, evidence references, contradictory evidence if applicable, uncertainty or blocker status, and the exact downstream components permitted to consume the result.

Failure modes and safeguards

Typical failure modes include schema drift, hidden manual edits, post-cutoff leakage, model misuse, confirmation bias, source echo, silent scope widening and irreproducible computation. FCEA treats these as release-blocking

when they affect a mandatory dependency. The default corrective action is downgrade, rerun or INCONCLUSIVE rather than forced promotion.

L21 - Proof Packet and Reproduction Package

Generate a self-contained package for every material verdict.

Mandatory controls

- Include evidence manifest, contradictory evidence, models, assumptions, counterfactuals, gate records, uncertainty and run manifest.
- A reviewer can trace every published statement to exact source versions.
- Package hashes permit tamper detection.
- Reproduction instructions specify environment and deterministic/stochastic tolerances.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

Required outputs

Outputs SHALL be machine-readable and human-readable. At minimum they include a result state, scope, method/version, run identifier, evidence references, contradictory evidence if applicable, uncertainty or blocker status, and the exact downstream components permitted to consume the result.

Failure modes and safeguards

Typical failure modes include schema drift, hidden manual edits, post-cutoff leakage, model misuse, confirmation bias, source echo, silent scope widening and irreproducible computation. FCEA treats these as release-blocking when they affect a mandatory dependency. The default corrective action is downgrade, rerun or INCONCLUSIVE rather than forced promotion.

L22 - Human Review and Governance

Introduce accountable review without hidden manual override.

Mandatory controls

- Review actions are immutable events.
- Reviewers may approve, downgrade, reject or request rerun.
- High-impact release policy may require independent/blinded review.
- Machine verdict and human disposition remain separately visible.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

Required outputs

Outputs SHALL be machine-readable and human-readable. At minimum they include a result state, scope, method/version, run identifier, evidence references, contradictory evidence if applicable, uncertainty or blocker status, and the exact downstream components permitted to consume the result.

Failure modes and safeguards

Typical failure modes include schema drift, hidden manual edits, post-cutoff leakage, model misuse, confirmation bias, source echo, silent scope widening and irreproducible computation. FCEA treats these as release-blocking when they affect a mandatory dependency. The default corrective action is downgrade, rerun or INCONCLUSIVE rather than forced promotion.

L23 - Security, Privacy and Production Operations

Protect evidence, users, secrets and reproducibility across acquisition, analytics and publication zones.

Mandatory controls

- Least privilege, MFA/SSO, encryption and network segmentation are baseline controls.
- Raw evidence cannot be silently mutated through analytical APIs.
- Backups and restore tests are release requirements.
- Privacy minimization and lawful-use policies are adapter/jurisdiction configurations.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

Required outputs

Outputs SHALL be machine-readable and human-readable. At minimum they include a result state, scope, method/version, run identifier, evidence references, contradictory evidence if applicable, uncertainty or blocker status, and the exact downstream components permitted to consume the result.

Failure modes and safeguards

Typical failure modes include schema drift, hidden manual edits, post-cutoff leakage, model misuse, confirmation bias, source echo, silent scope widening and irreproducible computation. FCEA treats these as release-blocking when they affect a mandatory dependency. The default corrective action is downgrade, rerun or INCONCLUSIVE rather than forced promotion.

L24 - Verification, Validation, Release and Publication

Define the final technical and scientific release gates.

Mandatory controls

- Run unit, property, integration, synthetic, historical, negative-control, blind-holdout and reproduction tests.
- Critical invariant failures block release.
- Release manifests freeze code, data and configuration identifiers.
- Corrections create new versioned records rather than rewriting historical outputs.

Required inputs

The layer SHALL declare all upstream object identifiers, schema versions, validity interval, missing-data policy, relevant uncertainty fields, and any adapter-specific preconditions. Any input used in a material output must be traceable to a registered source, model or prior gate record.

Required outputs

Outputs SHALL be machine-readable and human-readable. At minimum they include a result state, scope, method/version, run identifier, evidence references, contradictory evidence if applicable, uncertainty or blocker status, and the exact downstream components permitted to consume the result.

Failure modes and safeguards

Typical failure modes include schema drift, hidden manual edits, post-cutoff leakage, model misuse, confirmation bias, source echo, silent scope widening and irreproducible computation. FCEA treats these as release-blocking when they affect a mandatory dependency. The default corrective action is downgrade, rerun or INCONCLUSIVE rather than forced promotion.

Release = PASS only if all critical technical and scientific release gates PASS

PART IV - Mathematical Contracts and Algorithms

30. Status Lattices and Promotion Rules

FCEA deliberately avoids one global numeric score. Different dimensions have different semantics. The control plane uses explicit states and partial orders.

Axis	Illustrative order / relationship
Execution	NOT_RUN -> RUNNING -> COMPLETE;

	BLOCKED/ERROR/NOT_IMPLEMENTED are non-promotable terminal control states for the affected route.
Validity	VALID is required for scientific promotion; OUTSIDE_DOMAIN, MODEL_INVALID, PROVENANCE_FAILURE and IDENTIFICATION_FAILURE block interpretation.
Evidence	UNRESOLVED and INCONCLUSIVE cannot be silently promoted; SUPPORTED is weaker than CERTIFIED_WITHIN_SCOPE; FALSIFIED terminates the supported route.
Scope	Narrow scopes may widen only with a named certificate. A broad negative claim requires full-domain coverage appropriate to claim type.

30.1 Multi-route claims

$$\text{Eligible}(C) = \text{OR}_{\{r \text{ in Routes}(C)\}} \text{AND}_{\{d \text{ in } r\}} \text{Pass}(d)$$

If one route fails, another independently valid route may still establish the claim. The proof packet must name the route used; evidence from logically incompatible routes is not averaged into a synthetic score.

31. Evidence Independence

Let evidence items $E_1 \dots E_n$ originate from latent primary sources $P_1 \dots P_k$. Define dependency clusters using explicit citations, metadata, copied text fingerprints, acquisition lineage and domain-specific dependency rules.

$$N_{\text{eff}} = \sum_{j=1..k} w_j, \quad 0 < w_j \leq 1$$

Weights w_j are not universal truth weights. They are a conservative accounting device for independent corroboration and MUST be adapter-calibrated if used quantitatively. A safe default is to report cluster counts without converting them into probabilities.

32. Counterfactual Ensemble

$$C = \{CF_1, CF_2, \dots, CF_m\}$$

For estimand θ , compute θ_j under each plausible counterfactual CF_j and report robustness over the ensemble. Useful summaries include the range, distribution and sign stability, but the full model-specific outputs remain available.

$$\text{SignStability} = (1/m) * \sum_j 1[\text{sign}(\theta_j) = \text{sign}(\theta_{\text{ref}})]$$

SignStability is a descriptive robustness statistic, not a causal probability.

33. Causal Estimands

FCEA requires the estimand before the estimator. Common examples include ATE, ATT and time-indexed intervention effects.

$$\text{ATE} = E[Y(1) - Y(0)]$$

$$\text{ATT} = E[Y(1) - Y(0) \mid X=1]$$

$$\tau_{t,t} = E[Y_t \mid \text{do}(X=x_1)] - E[Y_t \mid \text{do}(X=x_0)]$$

The adapter must state the assumptions under which potential-outcome or structural-causal notation is meaningful. Mechanistic sciences may instead use parameter interventions, boundary-condition contrasts or validated simulation differences without pretending they are observational causal identification.

34. Difference-in-Differences Contract

$$\tau_{\text{DiD}} = (Y_{T,\text{post}} - Y_{T,\text{pre}}) - (Y_{C,\text{post}} - Y_{C,\text{pre}})$$

A conforming DiD adapter must document treatment timing, comparison units, anticipation, interference assumptions and pre-trend diagnostics. The formula alone is not sufficient for identification.

35. Instrumental Variable Contract

$$\text{LATE} = \text{Cov}(Z,Y) / \text{Cov}(Z,X)$$

This ratio is interpretable as a local average treatment effect only under the required relevance, exclusion, independence and monotonicity assumptions for the defined population. FCEA stores those assumptions as executable or reviewable validity predicates rather than hiding them in narrative text.

36. Bayesian and Frequentist Uncertainty

The core does not prefer one statistical paradigm. It requires the meaning of reported intervals to be explicit. A frequentist confidence interval is not interpreted as a posterior probability about a fixed parameter; a Bayesian credible interval requires a declared prior and posterior model.

37. Multiple Hypothesis Control

When many hypotheses, windows or subgroup analyses are searched, FCEA requires a declared multiplicity policy. Examples include family-wise error control or false-discovery-rate procedures. Exploratory discovery may use broader search, but confirmatory claims must state which multiplicity universe was controlled.

38. Closure Mathematics

Let a candidate path be $\pi = (v_0, e_1, v_1, \dots, e_k, v_k)$. Closure is type-dependent but always requires endpoint identity under the adapter semantics plus validity of all mandatory edges.

$$\text{CandidateClosed}(\pi) = \text{Identity_adapter}(v_0, v_k)$$

$$\text{CertifiedClosed}(\pi) = \text{CandidateClosed}(\pi) \text{ AND } \text{AND_i Valid}(e_i) \text{ AND Order}(\pi) \text{ AND ScopeOK}(\pi)$$

For weighted graph cycles, a geometric mean may summarize edge strength only when all edge weights share a calibrated, commensurable scale:

$$\text{CycleStrength}(C) = (\text{product}_{\{e \in C\} w_e})^{(1/|C|)}$$

If weight semantics differ across edges, FCEA prohibits multiplication and requires vector or edge-wise reporting.

39. Model Comparison

Model comparison is not model truth. A model can outperform alternatives while all models remain misspecified. FCEA records predictive performance, residual diagnostics, calibration, causal identification diagnostics and domain validity separately. The winning model does not inherit a stronger causal status than its identification assumptions support.

40. Evidence-to-Claim Join

Each claim C declares admissible evidence roles and proof routes. Let $\text{Sup}(C)$, $\text{Con}(C)$ and $\text{Neu}(C)$ be supporting, contradicting and neutral evidence sets. FCEA does not prescribe a universal scalar aggregation. Instead, the adapter defines a decision rule that must be transparent, preregisterable and falsifiable.

```
evaluate_claim(C):
  assert protocol_frozen(C)
  verify_scope(C)
  verify_required_models(C)
  verify_temporal_constraints(C)
  evaluate_supporting_evidence(C)
  evaluate_contradictory_evidence(C)
  run_counterfactuals_and_identification(C)
  run_falsification_tests(C)
  enforce_evidence_conservation(C)
  return strongest_permitted_scoped_verdict(C)
```

41. Prohibited Single-Score Shortcut

FCEA v1.0 prohibits a generic "truth score", "conspiracy score", "causality percentage" or "guilt probability" unless an adapter supplies an empirically calibrated labeled target, a validated probability model and explicit interpretation. Even then, the score cannot override hard provenance, validity or scope gates.

PART V - Federated Adapter Framework

42. Adapter Contract

```
AdapterSpec {
  adapter_id, version, domain, ontology_extensions,
  source_connectors, model_types, validity_predicates,
  time_semantics, counterfactual_semantics, causal_semantics,
  closure_semantics, gate_extensions, output_mapping,
  conformance_tests, legal_and_ethics_profile
}
```

Adapters translate domain knowledge into typed rules while the core retains universal controls. An adapter may impose stricter requirements but may not disable provenance, scope algebra, contradictory evidence handling, reproducibility, immutable audit events or the claim escalation firewall.

43. Adapter Conformance

Requirement	Conformance rule
Ontology	No core type may be silently redefined to mean something incompatible.
Validity	Every domain-specific model or estimator declares its applicability predicates.
Time	Temporal order and knowledge cutoff rules are explicit.
Causality	Causal claims identify an estimand and route appropriate to the domain.
Closure	Closure type and endpoint identity semantics are explicit.
Evidence	Evidence roles and minimum independent corroboration are documented.
Tests	Unit, integration, negative-control and benchmark tests are shipped with the adapter.
Governance	Prohibited inferences and lawful-use constraints are declared.

44. Example Adapter Families

Adapter family	Typical domain objects	Key extra gates
Scientific Theory	theories, equations, parameter regimes, experiments	equation/model validity, approximation control, benchmark reproduction
Financial / Market	prices, signals, positions, costs, liquidity	look-ahead control, transaction costs, slippage, holdout and regime stability
Policy Evaluation	interventions, populations, outcomes, institutions	identification, interference, parallel trends/positivity, counterfactual robustness
Public Health	exposure, treatment, outcome, confounders	measurement validity, selection bias, safety/ethics and causal identification
Climate Attribution	forcings, observations, models, fingerprints	model adequacy, covariance treatment, forcing uncertainty, attribution robustness
Supply Chain	suppliers, routes, inventories, dependencies	flow conservation, topology, substitution and cascade sensitivity
Integrity / Procurement	entities, contracts, owners, payments, influence	legal-state firewall, source independence, intent and causality separation
Physics / Causality	fields, equations, topology, trajectories, domains	on-shell/model validity, closure, stability and approximation validity

45. Market-Strategy Adapter Example

A market adapter demonstrates why FCEA separates evidence and causal validity from profitability metrics. A trading strategy may have a positive backtest and still fail because of look-ahead leakage, data survivorship, unmodeled transaction costs or unstable regime dependence.

$$R_{net,t} = R_{gross,t} - Costs_t - Slippage_t - Borrow_t$$

$$Sharpe = E[R_{excess}] / SD(R_{excess}) * \sqrt{A}$$

where A is the number of return periods per year under a consistent frequency convention. FCEA does not treat a high historical Sharpe ratio as a causal or forward guarantee. A conforming adapter requires temporal train/validation/test separation, cost modeling, turnover, drawdown, capacity and stability diagnostics.

PART VI - Data, API and Repository Standards

50. Authoritative Data Architecture

FCEA separates immutable evidence storage, normalized authoritative records, graph projections, analytical columnar workloads and search. Graph databases are optimized for traversal; they should not become the only authoritative store for source facts and audit history.

Store	Responsibility
Object store	Immutable raw sources, attachments and proof artifacts.
Relational database	Authoritative normalized objects, versions, scopes, gate records and audit events.
Graph database	Evidence, claim, model and dependency traversal projections.
Analytical engine	High-volume event, time-series and simulation tables.
Search index	Document and evidence retrieval with pointers to authoritative objects.
Artifact registry	Containers, software manifests, adapter packages and signed release artifacts.

51. Minimum Relational Schema

sources
source_versions
evidence_items
claims
claim_dependencies
evidence_claim_links
models
model_versions
implementations
assumptions
domains
scopes
events
temporal_relations
counterfactuals
causal_estimands
closure_candidates
tests
gate_runs
contradictions
source_dependencies
verdicts
proof_packets
reviews
run_manifests
audit_events

52. API Surface

Endpoint	Purpose
POST /v1/sources	Register/acquire a source; raw object remains immutable.
GET /v1/sources/{id}	Return metadata, versions and provenance.
POST /v1/evidence	Create a typed evidence item from a source-backed transform.

POST /v1/claims	Register a ClaimSpec and dependencies.
POST /v1/models	Register ModelSpec/ImplementationSpec versions.
POST /v1/runs	Execute a frozen protocol/model run.
GET /v1/runs/{id}	Return manifest, logs, gate outcomes and artifacts.
GET /v1/claims/{id}/proof	Return proof path and blockers.
POST /v1/reviews/{id}	Submit immutable reviewer disposition.
GET /v1/releases/{id}	Return signed release manifest and proof packets.

53. Repository Structure

fcea/
 core/
 types/ status/ scope/ invariants/
 schemas/
 sources/ provenance/ normalization/
 graphs/
 evidence/ claims/ models/ temporal/
 gates/ validity/ counterfactual/ causal/ closure/
 uncertainty/ falsification/ contradiction/ independence/ firewall/
 adapters/
 template/
 proof/ review/ api/ dashboard/
 security/ orchestration/ tests/ benchmarks/ infra/ docs/ manifests/

54. Run Manifest

run_id
timestamp
protocol_version
dataset_release
adapter_version
core_version
git_commit
container_digest
dependency_lock_hash
random_seed
parameters
hardware
environment
input_hashes
output_hashes
gate_results

A missing run manifest makes a model output NON_REPRODUCIBLE for material release even if the displayed value appears plausible.

PART VII - Verification, Validation, Security and Operations

60. Verification and Validation Ladder

Level	Test class	Purpose
V0	Schema validation	Reject malformed objects and incompatible versions before analytics.
V1	Unit tests	Verify individual formulas, transformations and predicates.
V2	Property/invariant tests	Check algebraic, dimensional, temporal and conservation properties.
V3	Integration tests	Validate multi-service object lineage and gate orchestration.

V4	Synthetic gold cases	Use cases with known expected truth and failure states.
V5	Historical benchmarks	Test reproducibility on frozen historical datasets without look-ahead.
V6	Negative controls / placebos	Measure false-positive behavior.
V7	Blind holdout	Separate selection/tuning from final evaluation.
V8	Adversarial / red-team	Search for prompt, data, model and governance failure modes.
V9	Fresh-environment reproduction	Re-run from manifests outside the development environment.
V10	Independent external replication	Test whether another team can reproduce the substantive result.

61. Mathematical Invariants

forall material claim c: Path(c, RawSource) != empty

forall historical object o: t_k(o) <= cutoff

Scope(output) <= Scope(valid_certificate)

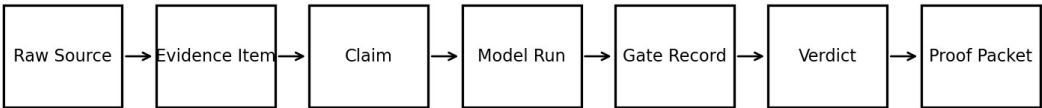
Strength(output) <= meet Strength(mandatory_dependencies)

DeterministicRun(inputs, version) = identical output

For explicitly stochastic components, exact byte equality is replaced by preregistered stochastic equivalence criteria, while random seeds and simulation design remain archived.

62. Security Architecture

Traceability Path



Every material conclusion must be reversible to the exact source version, transform, model run, assumptions and gate outcomes.

Figure 3. Material outputs remain traceable through the complete source-to-proof chain.

Zone	Purpose	Key control
Acquisition	Receive external/public/authorized source material	network isolation, malware scan, source metadata
Quarantine	Inspect untrusted files	content scanning and controlled parsers
Immutable evidence	Preserve raw bytes	write-once policy, encryption, access audit
Normalized data	Structured analytical records	schema validation, service identities
Restricted analytics	Sensitive/controlled computation	least privilege, secrets isolation, logging
Review	Human evaluation and adjudication	role separation and immutable review events
Publication/API	Release approved outputs	signed manifests, redaction/privacy checks

63. Mandatory Security Controls

- SSO/MFA for human users
- service-to-service identities

- TLS in transit
- encryption at rest
- secret manager
- least-privilege RBAC
- network segmentation
- immutable audit export
- container image signing
- SBOM and vulnerability scanning
- backup encryption
- restore testing
- data minimization and retention schedules
- redaction controls for public exports

64. Operational Runbook

1. Freeze protocol, adapter version, data cutoff and planned confirmatory methods.
2. Acquire and hash sources; quarantine untrusted content.
3. Run normalization, identity resolution and provenance validation.
4. Build evidence, claim, model and temporal graph releases.
5. Execute data-quality and validity gates.
6. Run model, counterfactual, identification and closure procedures.
7. Execute falsification, alternative explanation and independence tests.
8. Apply the Claim Escalation Firewall and generate the machine verdict.
9. Generate Proof Packet and full Run Manifest.
10. Route high-impact outputs to required human/blinded review.
11. Run V&V and fresh-environment reproduction.
12. Sign release manifest; publish only scoped reviewed conclusions.

65. Incident Response

13. Contain affected service or connector.
14. Preserve logs and evidence-state snapshots.
15. Identify data, model and claim impact.
16. Invalidate or quarantine affected outputs if required.
17. Restore a trusted version.
18. Rerun impacted analyses under the original frozen protocol when possible.
19. Issue a versioned correction record and preserve the superseded result for audit.

PART VIII - Deployment, Use Cases, Limitations and Governance

70. Implementation Roadmap

Phase	Deliverable	Exit criterion
0	Protocol and object schemas	Core types, status axes and scope rules approved.
1	Immutable source/provenance subsystem	Source-to-derived lineage verified on synthetic corpus.
2	Evidence/Claim/Model/Temporal graphs	Cross-graph joins pass integration tests.
3	Scope, validity and Gate Engine	Mandatory blockers and widening rules pass negative controls.
4	Counterfactual/causal/closure framework	At least two adapters exercise distinct semantics.
5	Falsification/alternatives/independence	False-positive and source-echo tests validated.
6	Proof Packet/reproduction	Independent reviewer reproduces test verdict from package.
7	Security and production API	Threat model, backup restore and access tests pass.
8	Blind historical pilot	Frozen holdout completed without post-cutoff

		leakage.
9	External replication	Independent implementation reproduces defined benchmarks.
10	Production release	Signed conformance and V&V package approved.

71. Who Benefits and How

FCEA is most useful when the cost of an unsupported conclusion is high. Researchers gain explicit assumptions and reproducibility. Public institutions gain transparent decision boundaries. Auditors gain chain-of-custody and proof packets. Engineers gain failure containment and gate semantics. Analysts gain a disciplined distinction between association, mechanism and causation. Reviewers gain the ability to reproduce both positive and negative findings instead of inspecting only narrative reports.

72. Example Use Case - Scientific Hypothesis

A scientific adapter registers a theory/model, domain, predicted observations and falsification conditions. Measurements enter the Evidence Graph; predicted relationships enter the Model Graph; experiment chronology enters the Temporal Graph. The system checks validity, uncertainties and scope. A positive result may support only the tested parameter regime. A failed search cannot produce a universal negative unless the adapter supplies exhaustive coverage or an appropriate theorem/certificate.

73. Example Use Case - Public Policy

A policy intervention is decomposed into implementation, exposure, intermediate mechanisms and outcome claims. Counterfactuals and an estimand are preregistered. Temporal and identification gates prevent post-treatment variables or future knowledge from entering the analysis. Alternative explanations and placebo dates are tested. The output states the strongest supported causal effect within the studied population and period, plus explicit limits on transferability.

74. Example Use Case - Financial Research

A trading hypothesis is separated from its implementation and backtest. The Temporal Graph enforces point-in-time data availability. Transaction costs, slippage and survivorship are explicit. Hyperparameter selection is isolated from the frozen test period. Robustness is checked across regimes and cost assumptions. The final claim may be "historically supported under declared assumptions" rather than "profitable in the future."

75. Example Use Case - Supply Chain Resilience

A supply-chain adapter models producers, transport links, inventory, substitution and demand. Closure semantics may represent circular dependencies or feedback cascades, but graph cycles do not automatically imply causal failure. Flow-conservation and capacity constraints are validated. Counterfactual disruption scenarios quantify exposure while uncertainty is retained for unknown inventories or substitution elasticities.

76. Governance and Ethics

FCEA governance is designed around traceability, neutrality and bounded claims. The system must preserve plausible benign explanations, protect personal data, respect lawful-use constraints and separate analytical findings from legal judgments. A reviewer cannot silently overwrite an automated result; the machine verdict and human disposition remain separate records.

77. Limitations

- No architecture can recover evidence that does not exist or is legally inaccessible.
- Unobserved confounding can prevent causal identification even with large datasets.
- Poor domain adapters can encode incorrect scientific assumptions; conformance does not replace domain expertise.
- Counterfactual conclusions remain model-dependent when no experimental identification is available.
- Source integrity does not guarantee source truth; it guarantees that the analyzed object is the registered object.

- Reproducibility does not guarantee correctness; an incorrect method can be reproducibly wrong.
- Human review can reduce but not eliminate institutional or cognitive bias.
- Universal claims remain exceptionally difficult and normally require broad empirical coverage or a valid theorem/certificate.

78. Effectiveness Metrics

Metric	Definition / objective
Traceability coverage	Fraction of material output fields with complete path to source and transform.
Reproduction success	Fraction of frozen releases reproduced within declared deterministic/stochastic tolerance.
Gate escape rate	Rate at which a known critical failure reaches downstream release; target zero.
False-positive rate	Measured on negative controls / benchmark cases appropriate to adapter.
Source-independence correction	Difference between nominal and effective corroboration after dependency clustering.
Scope violation count	Number of attempted unsupported promotions blocked by the firewall.
Correction latency	Time from validated incident discovery to published versioned correction.
Reviewer agreement	Agreement and adjudication statistics for blinded high-impact reviews.

79. Final Architecture Statement

FCEA v1.0 defines a complete architectural contract for building evidence-grade analytical systems. Its central contribution is not a new statistical estimator or one domain-specific model. It is a federated control plane that makes provenance, claim decomposition, model validity, temporal integrity, scope, causal identification, falsification, contradictory evidence and reproducibility jointly enforceable. The intended outcome is a system that can say not only what is supported, but also why, under which assumptions, within which domain, from which evidence, and what stronger statements are prohibited.

PART IX - Detailed Engineering Specifications for L0-L24

80. Purpose of the Detailed Engineering Layer

This part converts the architectural layers into implementation contracts. Each layer is described in terms of service boundary, canonical state transitions, required records, deterministic checks, observability, acceptance criteria and inter-layer interfaces. The purpose is to make an implementation reviewable before production data are introduced.

80.1 Engineering Specification - L0 Protocol Freeze and System Boundary

Engineering intent

Fix the analytical question, hypotheses, null, alternatives, observation window, cutoff, data universe, primary estimands, thresholds, falsification criteria and amendment policy before confirmatory analysis.

Service boundary

Owens protocol versions, research questions, hypothesis families, cutoffs, preregistered thresholds and amendment history. It must not own analytical results.

State transition

DRAFT -> REVIEWED -> FROZEN -> SUPERSEDED. A FROZEN protocol is immutable; amendments create a new protocol identifier.

Acceptance criteria

No confirmatory run may start without a FROZEN protocol; all thresholds referenced by a confirmatory result resolve to the exact frozen version.

Observability

Track amendment frequency, exploratory-to-confirmatory transitions, cutoff changes and threshold changes.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

20. Resolve all referenced object and schema versions.
21. Validate structural types, units and mandatory fields.
22. Evaluate relevant temporal, provenance and scope constraints.
23. Execute the layer-specific algorithm or rule under a Run Manifest.
24. Persist outputs and diagnostics as append-only records.
25. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L0 input produces the expected scoped output.
- Negative structural test: malformed L0 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.
- Reproducibility test: the same frozen L0 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L0 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.2 Engineering Specification - L1 Typed Ontology and Semantic Contracts

Engineering intent

Define canonical entity, relation, claim, evidence and model types so that semantically different objects cannot be combined by convenience.

Service boundary

Owns protocol versions, research questions, hypothesis families, cutoffs, preregistered thresholds and amendment history. It must not own analytical results.

State transition

DRAFT -> REVIEWED -> FROZEN -> SUPERSEDED. A FROZEN protocol is immutable; amendments create a new protocol identifier.

Acceptance criteria

No confirmatory run may start without a FROZEN protocol; all thresholds referenced by a confirmatory result resolve to the exact frozen version.

Observability

Track amendment frequency, exploratory-to-confirmatory transitions, cutoff changes and threshold changes.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

26. Resolve all referenced object and schema versions.

27. Validate structural types, units and mandatory fields.
28. Evaluate relevant temporal, provenance and scope constraints.
29. Execute the layer-specific algorithm or rule under a Run Manifest.
30. Persist outputs and diagnostics as append-only records.
31. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L1 input produces the expected scoped output.
- Negative structural test: malformed L1 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.
- Reproducibility test: the same frozen L1 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L1 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.3 Engineering Specification - L2 Source Integrity and Immutable Acquisition

Engineering intent

Acquire sources into an immutable raw store with cryptographic identifiers and acquisition metadata.

Service boundary

Owns source identities, source versions, cryptographic digests, transform lineage and parent-child relationships.

State transition

ACQUIRED -> VERIFIED -> PARSED -> DERIVED. Raw bytes never transition to a new semantic state; derived objects are new records.

Acceptance criteria

Randomly sampled published fields must traverse to exact source bytes and transform versions with no undocumented hop.

Observability

Hash mismatch rate, parser error rate, lineage-break count, unresolved transform count and source freshness.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

32. Resolve all referenced object and schema versions.
33. Validate structural types, units and mandatory fields.
34. Evaluate relevant temporal, provenance and scope constraints.
35. Execute the layer-specific algorithm or rule under a Run Manifest.
36. Persist outputs and diagnostics as append-only records.
37. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L2 input produces the expected scoped output.
- Negative structural test: malformed L2 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.

- Reproducibility test: the same frozen L2 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L2 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.4 Engineering Specification - L3 Provenance and Chain-of-Custody Engine

Engineering intent

Maintain reversible lineage from material verdicts to raw evidence and transformations.

Service boundary

Owns source identities, source versions, cryptographic digests, transform lineage and parent-child relationships.

State transition

ACQUIRED -> VERIFIED -> PARSED -> DERIVED. Raw bytes never transition to a new semantic state; derived objects are new records.

Acceptance criteria

Randomly sampled published fields must traverse to exact source bytes and transform versions with no undocumented hop.

Observability

Hash mismatch rate, parser error rate, lineage-break count, unresolved transform count and source freshness.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

38. Resolve all referenced object and schema versions.
39. Validate structural types, units and mandatory fields.
40. Evaluate relevant temporal, provenance and scope constraints.
41. Execute the layer-specific algorithm or rule under a Run Manifest.
42. Persist outputs and diagnostics as append-only records.
43. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L3 input produces the expected scoped output.
- Negative structural test: malformed L3 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.
- Reproducibility test: the same frozen L3 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L3 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.5 Engineering Specification - L4 Normalization, Resolution and Data Quality

Engineering intent

Normalize units and schemas while preserving raw values; resolve identities and quantify ambiguity.

Service boundary

Owns source identities, source versions, cryptographic digests, transform lineage and parent-child relationships.

State transition

ACQUIRED -> VERIFIED -> PARSED -> DERIVED. Raw bytes never transition to a new semantic state; derived objects are new records.

Acceptance criteria

Randomly sampled published fields must traverse to exact source bytes and transform versions with no undocumented hop.

Observability

Hash mismatch rate, parser error rate, lineage-break count, unresolved transform count and source freshness.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

44. Resolve all referenced object and schema versions.
45. Validate structural types, units and mandatory fields.
46. Evaluate relevant temporal, provenance and scope constraints.
47. Execute the layer-specific algorithm or rule under a Run Manifest.
48. Persist outputs and diagnostics as append-only records.
49. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L4 input produces the expected scoped output.
- Negative structural test: malformed L4 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.
- Reproducibility test: the same frozen L4 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L4 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.6 Engineering Specification - L5 Evidence Graph

Engineering intent

Represent supporting, contradicting, neutral and dependent evidence as graph objects.

Service boundary

Owns typed nodes and edges for its graph family while authoritative object payloads remain in the core stores.

State transition

BUILDING -> VALIDATED -> FROZEN_RELEASE -> SUPERSEDED. Graph projections are reproducible from authoritative records.

Acceptance criteria

No dangling mandatory references, illegal endpoint types, duplicate deterministic IDs or forbidden edge semantics.

Observability

Node/edge counts by type, orphan rate, temporal-invalid edge rate, dependency-cluster size and graph-release digest.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

50. Resolve all referenced object and schema versions.
51. Validate structural types, units and mandatory fields.
52. Evaluate relevant temporal, provenance and scope constraints.
53. Execute the layer-specific algorithm or rule under a Run Manifest.
54. Persist outputs and diagnostics as append-only records.
55. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L5 input produces the expected scoped output.
- Negative structural test: malformed L5 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.
- Reproducibility test: the same frozen L5 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L5 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.7 Engineering Specification - L6 Claim Graph

Engineering intent

Decompose broad narratives into atomic, testable propositions and explicit dependency chains.

Service boundary

Owns typed nodes and edges for its graph family while authoritative object payloads remain in the core stores.

State transition

BUILDING -> VALIDATED -> FROZEN_RELEASE -> SUPERSEDED. Graph projections are reproducible from authoritative records.

Acceptance criteria

No dangling mandatory references, illegal endpoint types, duplicate deterministic IDs or forbidden edge semantics.

Observability

Node/edge counts by type, orphan rate, temporal-invalid edge rate, dependency-cluster size and graph-release digest.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

56. Resolve all referenced object and schema versions.
57. Validate structural types, units and mandatory fields.
58. Evaluate relevant temporal, provenance and scope constraints.
59. Execute the layer-specific algorithm or rule under a Run Manifest.
60. Persist outputs and diagnostics as append-only records.
61. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L6 input produces the expected scoped output.
- Negative structural test: malformed L6 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.
- Reproducibility test: the same frozen L6 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L6 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.8 Engineering Specification - L7 Model Graph

Engineering intent

Represent theoretical, statistical, causal, numerical and mechanistic models separately from their implementations.

Service boundary

Owns typed nodes and edges for its graph family while authoritative object payloads remain in the core stores.

State transition

BUILDING -> VALIDATED -> FROZEN_RELEASE -> SUPERSEDED. Graph projections are reproducible from authoritative records.

Acceptance criteria

No dangling mandatory references, illegal endpoint types, duplicate deterministic IDs or forbidden edge semantics.

Observability

Node/edge counts by type, orphan rate, temporal-invalid edge rate, dependency-cluster size and graph-release digest.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

62. Resolve all referenced object and schema versions.
63. Validate structural types, units and mandatory fields.
64. Evaluate relevant temporal, provenance and scope constraints.

65. Execute the layer-specific algorithm or rule under a Run Manifest.
66. Persist outputs and diagnostics as append-only records.
67. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L7 input produces the expected scoped output.
- Negative structural test: malformed L7 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.
- Reproducibility test: the same frozen L7 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L7 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked **BLOCKED** or **INVALID**, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.9 Engineering Specification - L8 Temporal Graph and Knowledge Cutoff

Engineering intent

Enforce temporal ordering, interval semantics and knowledge-time constraints.

Service boundary

Owns typed nodes and edges for its graph family while authoritative object payloads remain in the core stores.

State transition

BUILDING -> VALIDATED -> FROZEN_RELEASE -> SUPERSEDED. Graph projections are reproducible from authoritative records.

Acceptance criteria

No dangling mandatory references, illegal endpoint types, duplicate deterministic IDs or forbidden edge semantics.

Observability

Node/edge counts by type, orphan rate, temporal-invalid edge rate, dependency-cluster size and graph-release digest.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

68. Resolve all referenced object and schema versions.
69. Validate structural types, units and mandatory fields.
70. Evaluate relevant temporal, provenance and scope constraints.
71. Execute the layer-specific algorithm or rule under a Run Manifest.
72. Persist outputs and diagnostics as append-only records.
73. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L8 input produces the expected scoped output.
- Negative structural test: malformed L8 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.

- Reproducibility test: the same frozen L8 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L8 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.10 Engineering Specification - L9 Scope Algebra

Engineering intent

Attach every claim, test and verdict to a typed scope and prevent unsupported widening.

Service boundary

Owns typed nodes and edges for its graph family while authoritative object payloads remain in the core stores.

State transition

BUILDING -> VALIDATED -> FROZEN_RELEASE -> SUPERSEDED. Graph projections are reproducible from authoritative records.

Acceptance criteria

No dangling mandatory references, illegal endpoint types, duplicate deterministic IDs or forbidden edge semantics.

Observability

Node/edge counts by type, orphan rate, temporal-invalid edge rate, dependency-cluster size and graph-release digest.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

74. Resolve all referenced object and schema versions.
75. Validate structural types, units and mandatory fields.
76. Evaluate relevant temporal, provenance and scope constraints.
77. Execute the layer-specific algorithm or rule under a Run Manifest.
78. Persist outputs and diagnostics as append-only records.
79. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L9 input produces the expected scoped output.
- Negative structural test: malformed L9 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.
- Reproducibility test: the same frozen L9 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L9 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.11 Engineering Specification - L10 Validity and Assumption Engine

Engineering intent

Evaluate whether data, models and assumptions are valid for the declared domain before substantive interpretation.

Service boundary

Owns a defined analytical computation and its diagnostics; it does not own publication wording.

State transition

REGISTERED -> READY -> RUNNING -> COMPLETE/FAILED/BLOCKED. COMPLETE is not equivalent to scientific support.

Acceptance criteria

Formula/unit tests, synthetic cases, sensitivity tests and run-manifest completeness pass.

Observability

Runtime, convergence, failure codes, sensitivity range, numerical warnings and model-version use.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

80. Resolve all referenced object and schema versions.
81. Validate structural types, units and mandatory fields.
82. Evaluate relevant temporal, provenance and scope constraints.
83. Execute the layer-specific algorithm or rule under a Run Manifest.
84. Persist outputs and diagnostics as append-only records.
85. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L10 input produces the expected scoped output.
- Negative structural test: malformed L10 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.
- Reproducibility test: the same frozen L10 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L10 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.12 Engineering Specification - L11 Gate Engine

Engineering intent

Execute prerequisite gates and block downstream claims when required certificates fail or remain unresolved.

Service boundary

Owns a defined analytical computation and its diagnostics; it does not own publication wording.

State transition

REGISTERED -> READY -> RUNNING -> COMPLETE/FAILED/BLOCKED. COMPLETE is not equivalent to scientific support.

Acceptance criteria

Formula/unit tests, synthetic cases, sensitivity tests and run-manifest completeness pass.

Observability

Runtime, convergence, failure codes, sensitivity range, numerical warnings and model-version use.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

86. Resolve all referenced object and schema versions.
87. Validate structural types, units and mandatory fields.
88. Evaluate relevant temporal, provenance and scope constraints.
89. Execute the layer-specific algorithm or rule under a Run Manifest.
90. Persist outputs and diagnostics as append-only records.
91. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L11 input produces the expected scoped output.
- Negative structural test: malformed L11 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.
- Reproducibility test: the same frozen L11 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L11 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.13 Engineering Specification - L12 Counterfactual Engine

Engineering intent

Define relevant alternative worlds or baselines and estimate how conclusions change under them.

Service boundary

Owns a defined analytical computation and its diagnostics; it does not own publication wording.

State transition

REGISTERED -> READY -> RUNNING -> COMPLETE/FAILED/BLOCKED. COMPLETE is not equivalent to scientific support.

Acceptance criteria

Formula/unit tests, synthetic cases, sensitivity tests and run-manifest completeness pass.

Observability

Runtime, convergence, failure codes, sensitivity range, numerical warnings and model-version use.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

92. Resolve all referenced object and schema versions.

93. Validate structural types, units and mandatory fields.
94. Evaluate relevant temporal, provenance and scope constraints.
95. Execute the layer-specific algorithm or rule under a Run Manifest.
96. Persist outputs and diagnostics as append-only records.
97. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L12 input produces the expected scoped output.
- Negative structural test: malformed L12 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.
- Reproducibility test: the same frozen L12 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L12 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.14 Engineering Specification - L13 Causal Identification Engine

Engineering intent

Require a stated estimand and identification strategy for causal claims.

Service boundary

Owns a defined analytical computation and its diagnostics; it does not own publication wording.

State transition

REGISTERED -> READY -> RUNNING -> COMPLETE/FAILED/BLOCKED. COMPLETE is not equivalent to scientific support.

Acceptance criteria

Formula/unit tests, synthetic cases, sensitivity tests and run-manifest completeness pass.

Observability

Runtime, convergence, failure codes, sensitivity range, numerical warnings and model-version use.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

98. Resolve all referenced object and schema versions.
99. Validate structural types, units and mandatory fields.
100. Evaluate relevant temporal, provenance and scope constraints.
101. Execute the layer-specific algorithm or rule under a Run Manifest.
102. Persist outputs and diagnostics as append-only records.
103. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L13 input produces the expected scoped output.
- Negative structural test: malformed L13 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.

- Reproducibility test: the same frozen L13 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L13 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.15 Engineering Specification - L14 Closure Engine

Engineering intent

Verify global path closure, feedback mechanisms or domain-specific closed structures without conflating different closure semantics.

Service boundary

Owens a defined analytical computation and its diagnostics; it does not own publication wording.

State transition

REGISTERED -> READY -> RUNNING -> COMPLETE/FAILED/BLOCKED. COMPLETE is not equivalent to scientific support.

Acceptance criteria

Formula/unit tests, synthetic cases, sensitivity tests and run-manifest completeness pass.

Observability

Runtime, convergence, failure codes, sensitivity range, numerical warnings and model-version use.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

104. Resolve all referenced object and schema versions.
105. Validate structural types, units and mandatory fields.
106. Evaluate relevant temporal, provenance and scope constraints.
107. Execute the layer-specific algorithm or rule under a Run Manifest.
108. Persist outputs and diagnostics as append-only records.
109. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L14 input produces the expected scoped output.
- Negative structural test: malformed L14 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.
- Reproducibility test: the same frozen L14 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L14 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.16 Engineering Specification - L15 Uncertainty Engine

Engineering intent

Represent and propagate data, measurement, parameter, model, counterfactual, structural and scope uncertainty.

Service boundary

Owns a defined analytical computation and its diagnostics; it does not own publication wording.

State transition

REGISTERED -> READY -> RUNNING -> COMPLETE/FAILED/BLOCKED. COMPLETE is not equivalent to scientific support.

Acceptance criteria

Formula/unit tests, synthetic cases, sensitivity tests and run-manifest completeness pass.

Observability

Runtime, convergence, failure codes, sensitivity range, numerical warnings and model-version use.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

110. Resolve all referenced object and schema versions.
111. Validate structural types, units and mandatory fields.
112. Evaluate relevant temporal, provenance and scope constraints.
113. Execute the layer-specific algorithm or rule under a Run Manifest.
114. Persist outputs and diagnostics as append-only records.
115. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L15 input produces the expected scoped output.
- Negative structural test: malformed L15 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.
- Reproducibility test: the same frozen L15 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L15 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.17 Engineering Specification - L16 Falsification Engine

Engineering intent

Attempt to disprove or weaken hypotheses through negative controls, placebos, adversarial tests and competing predictions.

Service boundary

Owns a defined analytical computation and its diagnostics; it does not own publication wording.

State transition

REGISTERED -> READY -> RUNNING -> COMPLETE/FAILED/BLOCKED. COMPLETE is not equivalent to scientific support.

Acceptance criteria

Formula/unit tests, synthetic cases, sensitivity tests and run-manifest completeness pass.

Observability

Runtime, convergence, failure codes, sensitivity range, numerical warnings and model-version use.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

116. Resolve all referenced object and schema versions.
117. Validate structural types, units and mandatory fields.
118. Evaluate relevant temporal, provenance and scope constraints.
119. Execute the layer-specific algorithm or rule under a Run Manifest.
120. Persist outputs and diagnostics as append-only records.
121. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L16 input produces the expected scoped output.
- Negative structural test: malformed L16 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.
- Reproducibility test: the same frozen L16 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L16 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.18 Engineering Specification - L17 Contradiction and Alternative-Explanation Engine

Engineering intent

Collect and evaluate evidence and models that could explain the observations without the target hypothesis.

Service boundary

Owns a defined analytical computation and its diagnostics; it does not own publication wording.

State transition

REGISTERED -> READY -> RUNNING -> COMPLETE/FAILED/BLOCKED. COMPLETE is not equivalent to scientific support.

Acceptance criteria

Formula/unit tests, synthetic cases, sensitivity tests and run-manifest completeness pass.

Observability

Runtime, convergence, failure codes, sensitivity range, numerical warnings and model-version use.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

122. Resolve all referenced object and schema versions.

- 123. Validate structural types, units and mandatory fields.
- 124. Evaluate relevant temporal, provenance and scope constraints.
- 125. Execute the layer-specific algorithm or rule under a Run Manifest.
- 126. Persist outputs and diagnostics as append-only records.
- 127. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L17 input produces the expected scoped output.
- Negative structural test: malformed L17 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.
- Reproducibility test: the same frozen L17 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L17 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.19 Engineering Specification - L18 Evidence Independence Engine

Engineering intent

Estimate effective corroboration after accounting for shared upstream sources and duplicated reporting.

Service boundary

Owns source identities, source versions, cryptographic digests, transform lineage and parent-child relationships.

State transition

ACQUIRED -> VERIFIED -> PARSED -> DERIVED. Raw bytes never transition to a new semantic state; derived objects are new records.

Acceptance criteria

Randomly sampled published fields must traverse to exact source bytes and transform versions with no undocumented hop.

Observability

Hash mismatch rate, parser error rate, lineage-break count, unresolved transform count and source freshness.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

- 128. Resolve all referenced object and schema versions.
- 129. Validate structural types, units and mandatory fields.
- 130. Evaluate relevant temporal, provenance and scope constraints.
- 131. Execute the layer-specific algorithm or rule under a Run Manifest.
- 132. Persist outputs and diagnostics as append-only records.
- 133. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L18 input produces the expected scoped output.
- Negative structural test: malformed L18 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.

- Reproducibility test: the same frozen L18 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L18 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.20 Engineering Specification - L19 Claim Escalation Firewall

Engineering intent

Prevent prohibited inferential jumps and enforce the Evidence Conservation Principle.

Service boundary

Owens a defined analytical computation and its diagnostics; it does not own publication wording.

State transition

REGISTERED -> READY -> RUNNING -> COMPLETE/FAILED/BLOCKED. COMPLETE is not equivalent to scientific support.

Acceptance criteria

Formula/unit tests, synthetic cases, sensitivity tests and run-manifest completeness pass.

Observability

Runtime, convergence, failure codes, sensitivity range, numerical warnings and model-version use.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

134. Resolve all referenced object and schema versions.
135. Validate structural types, units and mandatory fields.
136. Evaluate relevant temporal, provenance and scope constraints.
137. Execute the layer-specific algorithm or rule under a Run Manifest.
138. Persist outputs and diagnostics as append-only records.
139. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L19 input produces the expected scoped output.
- Negative structural test: malformed L19 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.
- Reproducibility test: the same frozen L19 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L19 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.21 Engineering Specification - L20 Federated Adapter Framework

Engineering intent

Isolate domain-specific ontology, validity, causal and closure semantics while preserving core invariants.

Service boundary

Owns a defined analytical computation and its diagnostics; it does not own publication wording.

State transition

REGISTERED -> READY -> RUNNING -> COMPLETE/FAILED/BLOCKED. COMPLETE is not equivalent to scientific support.

Acceptance criteria

Formula/unit tests, synthetic cases, sensitivity tests and run-manifest completeness pass.

Observability

Runtime, convergence, failure codes, sensitivity range, numerical warnings and model-version use.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

140. Resolve all referenced object and schema versions.
141. Validate structural types, units and mandatory fields.
142. Evaluate relevant temporal, provenance and scope constraints.
143. Execute the layer-specific algorithm or rule under a Run Manifest.
144. Persist outputs and diagnostics as append-only records.
145. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L20 input produces the expected scoped output.
- Negative structural test: malformed L20 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.
- Reproducibility test: the same frozen L20 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L20 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.22 Engineering Specification - L21 Proof Packet and Reproduction Package

Engineering intent

Generate a self-contained package for every material verdict.

Service boundary

Owns release disposition, reviewer events, access controls and compliance evidence; it cannot silently alter machine analytical records.

State transition

PENDING_REVIEW -> APPROVED/DOWNGRADED/REJECTED/RERUN_REQUIRED.

Acceptance criteria

Every material review is attributable, timestamped, reasoned and immutable; conflicts of interest are recorded where policy requires.

Observability

Queue age, downgrade rate, reviewer agreement, security events, correction latency and release-blocker counts.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

146. Resolve all referenced object and schema versions.
147. Validate structural types, units and mandatory fields.
148. Evaluate relevant temporal, provenance and scope constraints.
149. Execute the layer-specific algorithm or rule under a Run Manifest.
150. Persist outputs and diagnostics as append-only records.
151. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L21 input produces the expected scoped output.
- Negative structural test: malformed L21 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.
- Reproducibility test: the same frozen L21 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L21 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.23 Engineering Specification - L22 Human Review and Governance

Engineering intent

Introduce accountable review without hidden manual override.

Service boundary

Owns release disposition, reviewer events, access controls and compliance evidence; it cannot silently alter machine analytical records.

State transition

PENDING_REVIEW -> APPROVED/DOWNGRADED/REJECTED/RERUN_REQUIRED.

Acceptance criteria

Every material review is attributable, timestamped, reasoned and immutable; conflicts of interest are recorded where policy requires.

Observability

Queue age, downgrade rate, reviewer agreement, security events, correction latency and release-blocker counts.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

152. Resolve all referenced object and schema versions.
153. Validate structural types, units and mandatory fields.
154. Evaluate relevant temporal, provenance and scope constraints.
155. Execute the layer-specific algorithm or rule under a Run Manifest.
156. Persist outputs and diagnostics as append-only records.

157. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L22 input produces the expected scoped output.
- Negative structural test: malformed L22 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.
- Reproducibility test: the same frozen L22 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L22 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.24 Engineering Specification - L23 Security, Privacy and Production Operations

Engineering intent

Protect evidence, users, secrets and reproducibility across acquisition, analytics and publication zones.

Service boundary

Owns release disposition, reviewer events, access controls and compliance evidence; it cannot silently alter machine analytical records.

State transition

PENDING_REVIEW -> APPROVED/DOWNGRADED/REJECTED/RERUN_REQUIRED.

Acceptance criteria

Every material review is attributable, timestamped, reasoned and immutable; conflicts of interest are recorded where policy requires.

Observability

Queue age, downgrade rate, reviewer agreement, security events, correction latency and release-blocker counts.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

158. Resolve all referenced object and schema versions.
159. Validate structural types, units and mandatory fields.
160. Evaluate relevant temporal, provenance and scope constraints.
161. Execute the layer-specific algorithm or rule under a Run Manifest.
162. Persist outputs and diagnostics as append-only records.
163. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L23 input produces the expected scoped output.
- Negative structural test: malformed L23 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.
- Reproducibility test: the same frozen L23 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L23 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

80.25 Engineering Specification - L24 Verification, Validation, Release and Publication

Engineering intent

Define the final technical and scientific release gates.

Service boundary

Owns release disposition, reviewer events, access controls and compliance evidence; it cannot silently alter machine analytical records.

State transition

PENDING_REVIEW -> APPROVED/DOWNGRADED/REJECTED/RERUN_REQUIRED.

Acceptance criteria

Every material review is attributable, timestamped, reasoned and immutable; conflicts of interest are recorded where policy requires.

Observability

Queue age, downgrade rate, reviewer agreement, security events, correction latency and release-blocker counts.

Interface contract

The layer consumes only versioned upstream objects. Every write emits a new object or an immutable event. The layer must return a structured status, scope, run identifier and blocker list. A downstream service must be able to reject incompatible schema or adapter versions before computation begins.

Deterministic validation sequence

164. Resolve all referenced object and schema versions.
165. Validate structural types, units and mandatory fields.
166. Evaluate relevant temporal, provenance and scope constraints.
167. Execute the layer-specific algorithm or rule under a Run Manifest.
168. Persist outputs and diagnostics as append-only records.
169. Execute negative and invariant checks before exposing the result downstream.

Minimum test set

- Positive test: a valid L24 input produces the expected scoped output.
- Negative structural test: malformed L24 input is rejected before computation.
- Boundary test: a value exactly at a validity or scope boundary is handled according to the declared inclusive/exclusive rule.
- Reproducibility test: the same frozen L24 inputs and implementation produce the same deterministic result, or the preregistered stochastic equivalence result.
- Propagation test: an L24 blocker prevents dependent claim promotion.

Operational failure response

A critical failure does not trigger best-effort continuation. The affected object is marked BLOCKED or INVALID, dependent outputs are quarantined, and the audit stream records the exact failure code. If a later correction is made, the corrected run receives a new identifier and the old record remains traceable.

81. Detailed Gate Engine Algorithm

```
def evaluate_gate_chain(claim, required_gates, context):
    records = []
    for gate in topological_order(required_gates):
        if any(parent.outcome != "PASS" for parent in records if parent.id in gate.parents):
            rec = GateRecord(gate, outcome="BLOCKED", blockers="upstream gate")
```

```

else:
    rec = gate.execute(claim, context)
    persist_immutable(rec)
    records.append(rec)
return records

```

The control graph for gates MUST be acyclic even when the domain model contains feedback loops. This prevents circular certification in which two unverified conclusions validate one another.

GateGraph = (V_G, E_G), GateGraph must be a DAG

82. Scope-Widening Algorithm

```

def widen_scope(claim, target_scope, certificate):
    if target_scope <= claim.scope:
        return claim.with_scope(target_scope)
    if certificate is None:
        raise ScopeViolation("widening certificate required")
    if not certificate.valid_for(claim, target_scope):
        raise ScopeViolation("certificate does not cover target scope")
    return claim.with_scope(target_scope, widening_certificate=certificate.id)

```

A widening certificate is itself evidence-bearing and versioned. Examples include an external replication set spanning the larger population, a theorem whose hypotheses are verified over the larger domain, or a statistically valid sampling/generalization design. A narrative assertion that a result is "likely general" is not a certificate.

83. Temporal Leakage Detection

$$\text{Leak}(o, t_c) = 1[t_k(o) > t_c]$$

The leakage detector traverses all evidence, features, labels and model-selection inputs used by a historical or prospective run. The test includes direct sources and derived objects. If a derived feature was computed using any parent object unavailable at the cutoff, the feature is also inadmissible.

Admissible(derived) = AND_{p in Parents(derived)} Admissible(p)

84. Evidence Dependency Clustering

Evidence dependence is represented through explicit citations, provenance parents, text/content similarity, common datasets and domain-specific source lineage. Clustering may use connected components on high-confidence dependency edges or probabilistic community models, but uncertain dependency must remain visible rather than being forced into independence.

G_dep = (E, D), where (e_i, e_j) in D means evidence items share a material upstream dependency

For a conservative release, the system reports both nominal and dependency-adjusted corroboration counts. Quantitative weighting is optional and must be separately calibrated.

85. Falsification Scheduling

Falsification tests are scheduled before the final confirmatory verdict where technically possible. Negative controls, placebo windows and synthetic falsifiers are bound to the hypothesis version, preventing selective omission after an inconvenient result.

```

Hypothesis H
-> required_falsifiers: [F1, F2, F3]
-> execute all required F_i
-> any critical falsifier PASS-against-H => downgrade/falsify route
-> unresolved falsifier => cap claim strength
-> only then evaluate final promotion

```

86. Proof-Route Algebra

Many real claims have more than one logically valid proof route. FCEA represents each route as a conjunction of mandatory certificates; routes are then combined disjunctively.

$$\text{Proof}(C) = R_1 \text{ OR } R_2 \text{ OR } \dots \text{ OR } R_m$$

$$R_j = D_{\{1\}} \text{ AND } D_{\{2\}} \text{ AND } \dots \text{ AND } D_{\{k_j\}}$$

This is superior to additive scoring when routes have different semantics. A missing dependency in one route cannot be numerically compensated by unrelated evidence from another route.

87. Contradiction Severity and Blocking

Contradictions are classified by whether they attack source authenticity, measurement, model assumptions, causal identification, mechanism, temporal order or scope. Severity is not a moral label; it determines whether the contradiction blocks promotion, requires sensitivity analysis or is merely disclosed.

Contradiction class	Typical effect
Authenticity/provenance	Blocks use of the affected evidence until resolved.
Measurement conflict	Requires reconciliation, uncertainty widening or competing measurement route.
Model-assumption conflict	Blocks model-based interpretation in the affected domain.
Temporal conflict	Blocks causal ordering when material.
Identification conflict	Downgrades causal claim to association/unresolved.
Scope conflict	Restricts the verdict to the intersection of validated scopes.
Minor contextual conflict	Disclosed; may not block if not material to the claim.

88. Reproduction Tolerances

Deterministic components should reproduce bit-for-bit where environment and numeric backend permit. When exact equality is not portable, the implementation defines deterministic numerical tolerances tied to conditioning and algorithm precision. Stochastic components require distributional or summary-statistic tolerances preregistered before rerun.

$$|x_{\text{repro}} - x_{\text{ref}}| \leq \text{atol} + \text{rtol} * |x_{\text{ref}}|$$

The values atol and rtol are model-specific engineering tolerances, not generic defaults. They must be justified by numeric precision, conditioning and the scientific scale of the result.

89. Example Full Claim Lifecycle

170. Register hypothesis H and atomic terminal claim C under a frozen protocol.
171. Acquire source objects S1...Sn and verify cryptographic integrity.
172. Create evidence items with valid-time, knowledge-time, transforms and uncertainty.
173. Build Claim Graph dependencies and record competing claims.
174. Register ModelSpec and ImplementationSpec, including assumptions and validity predicates.
175. Freeze dataset and graph releases; execute provenance and temporal gates.
176. Run the estimator or mechanistic computation and persist a Run Manifest.
177. Run counterfactual, identification and closure procedures required by the claim type.
178. Execute sensitivity, negative controls, placebos and competing explanations.
179. Construct source-dependency clusters and recalculate corroboration semantics.
180. Apply Evidence Conservation and Scope Algebra; determine the strongest permitted verdict.
181. Generate Proof Packet, run reproduction, obtain required human review and sign release.

90. Production Sizing and Performance Principles

FCEA is not tied to a specific cloud or database vendor. Performance design should follow workload separation. Raw evidence ingestion is throughput- and integrity-oriented; graph traversal is relationship-oriented; time-series and simulation workloads are columnar/analytical; review and publication need low-latency indexed retrieval. Scaling one store for every workload is discouraged because it reduces auditability and operational clarity.

Workload	Primary scaling concern	Recommended design principle
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Raw evidence	object count, file size, ingest burst	immutable object storage with content-addressed identities
Normalized records	transaction consistency, version history	relational authoritative store with migrations
Graph traversal	high-degree nodes, multi-hop paths	graph projection with bounded queries and cached releases
Time series	large scans, joins, simulations	columnar engine and partitioned historical releases
Search	document retrieval	search index pointing to authoritative source IDs
Proof packets	artifact immutability	signed artifact store and release registry

91. Availability and Disaster-Recovery Requirements

Production deployments must define Recovery Point Objective (RPO) and Recovery Time Objective (RTO) by data class. Raw evidence and audit history generally require stricter durability than rebuildable graph projections. A graph projection can be regenerated from authoritative stores; a lost raw source or audit event may destroy traceability.

92. Change Management

Any change to formulas, schemas, adapter semantics, thresholds, causal identification logic or gate policy is versioned. A release that changes interpretation must not silently reuse prior verdict identifiers. Historical results remain linked to the exact old software and protocol versions that produced them.

93. Documentation Requirements

- Architecture Decision Records for material design choices.
- Data dictionaries for every published field.
- Adapter scientific rationale and validity domain.
- Runbooks for ingestion, analysis, release and incident response.
- Reproduction instructions for benchmark and production releases.
- Known limitations and prohibited claims.
- Change log mapping each release to requirement and regression-test changes.

94. Staffing and Role Separation

Role	Responsibility	Separation concern
System architect	core contracts, integration, scope and gate policy	should not unilaterally approve domain truth claims
Data engineer	ingestion, normalization, quality and lineage	cannot silently alter analytical thresholds
Domain scientist/analyst	models, assumptions, estimands and interpretation	cannot bypass provenance/security gates
Causal-method reviewer	identification and counterfactual review	independent from result-driven tuning where feasible
Security engineer	access, secrets, network and supply-chain controls	separate production privileges from analysts
Independent reviewer	proof-packet and conclusion review	blind to preferred outcome when policy requires
Release owner	final conformance and publication decision	must preserve machine and reviewer records

95. Deployment Profiles

Profile	Use	Minimum components
Research Local	single-team experiments	immutable local source store, relational core, graph engine, manifests and test suite
Institutional Isolated	sensitive oversight/research	segmented network, internal artifact registry, RBAC/MFA, review zone, offline releases
Controlled Cloud	scalable multi-team deployment	managed object/relational stores, isolated compute, signed images, centralized audit and secret manager
Publication Replica	public reproducibility	frozen non-sensitive dataset release, open

		code/artifacts, proof packets and signed manifest
--	--	---

96. Example Research Protocol YAML

```

protocol_id: FCEA-DEMO-001
mode: confirmatory
research_question: "Does X cause Y within population P during interval T?"
knowledge_cutoff: 2026-01-01T00:00:00Z
hypotheses:
  - id: H0
    statement: "No material causal effect under estimand E"
  - id: H1
    statement: "A material causal effect exists under estimand E"
primary_estimand: ATE
required_gates: [G0,G2,G3,G5,G6,G7,G8,G9,G11,G12,G13,G14,G15]
missing_data_policy: preregistered
multiple_testing_policy: declared
release_rule: "No causal wording unless G8 and G9 PASS"

```

97. Example Verdict JSON

```

{
  "verdict_id": "VRD-001",
  "claim_id": "CLM-001",
  "execution_state": "COMPLETE",
  "scientific_state": "SUPPORTED",
  "validity_state": "VALID",
  "scope_ref": "SCP-001",
  "supporting_evidence": ["E-1", "E-4"],
  "contradictory_evidence": ["E-9"],
  "missing_evidence": ["E-required-12"],
  "alternative_explanations": ["ALT-2"],
  "prohibited_claims": [
    "Universal generalization beyond SCP-001",
    "Intent attribution"
  ],
  "proof_packet_ref": "PP-001"
}

```

98. Scientific Interpretation Rules

Observed system state	Permitted language	Prohibited escalation
Authentic observation only	"Observed", "recorded", "measured"	"caused", "proved mechanism"
Robust association	"associated with", "predictive within data"	"causes" without identification
Identified causal estimate	"estimated causal effect under assumptions"	universal claim outside scope
Mechanism supported	"mechanism supported in declared domain"	intent/motive unless separately evidenced
Failed falsification attempt	"not falsified by specified tests"	"proved true"
Inconclusive evidence	"inconclusive" or "unresolved"	forced PASS/FAIL
Validated local theorem/certificate	scope-limited certification language	broader theory/population conclusion without widening certificate

99. Implementation Readiness Statement

FCEA v1.0 is implementation-ready as an architectural and methodological standard: the core object model, graph separation, temporal and scope rules, gate semantics, counterfactual and causal contracts, falsification requirements, adapter interface, proof packet, run manifest, V&V ladder, security zones, operational controls and release logic are specified. A real deployment still requires software implementation, domain-adapter validation,

legal/privacy configuration, representative benchmark data, performance engineering and independent verification before production conclusions are issued.

APPENDIX A - Normative Data Contracts

SourceSpec

source_id: string
version: string
origin_uri_or_reference: string
acquired_at: datetime
hash_algorithm: SHA-256
hash_value: hex
legal_basis: string
raw_object_uri: string
parser_status: enum
provenance_parent: null|string

EvidenceItem

evidence_id: string
source_version_ref: string
claim_roles: list
value: typed
units: string|null
valid_time: interval|null
knowledge_time: datetime
uncertainty: object
transform_ref: string
quality_state: enum

ClaimSpec

claim_id: string
proposition: string
claim_type: enum
scope_ref: string
mandatory_dependencies: list
optional_dependencies: list
required_gates: list
prohibited_inferences: list
proof_routes: list
status: object

ModelSpec

model_id: string
version: string
model_type: string
equations_or_rules: artifact_ref
parameters: object
assumptions: list
validity_predicates: list
estimand_ref: string|null
implementation_refs: list
provenance: list

GateRecord

gate_run_id: string
 gate_id: string
 input_digests: list
 method_version: string
 outcome: PASS | FAIL | INCONCLUSIVE | BLOCKED | NOT_IMPLEMENTED
 blockers: list
 run_manifest_ref: string
 review_ref: string | null

VerdictRecord

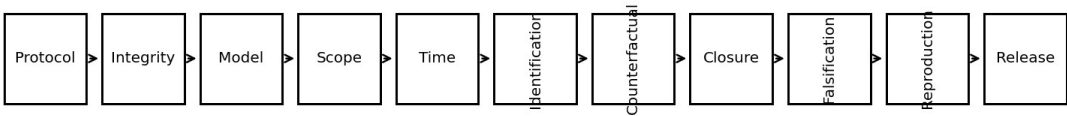
verdict_id: string
 claim_ref: string
 execution_state: enum
 scientific_state: enum
 validity_state: enum
 scope_ref: string
 supporting_evidence: list
 contradictory_evidence: list
 missing_evidence: list
 alternative_explanations: list
 prohibited_claims: list
 proof_packet_ref: string

APPENDIX B - Gate Catalogue

Gate	Name	Normative purpose
G0	Protocol Freeze	Protocol, hypotheses, thresholds and cutoff are immutable for the run.
G1	Structural/Schema	All objects conform to registered schemas and compatible versions.
G2	Source Integrity	Required raw sources are present, hashed and not corrupted.
G3	Provenance	Material inputs and outputs have complete lineage.
G4	Data Quality	Critical fields, units, duplicates and identity resolutions meet thresholds.
G5	Model Validity	Model assumptions and domain predicates are satisfied.
G6	Scope Validity	Claim scope does not exceed its supporting certificate.
G7	Temporal Consistency	Required event order and knowledge cutoff are valid.
G8	Identification	Causal estimand is identified under declared assumptions.
G9	Counterfactual	Relevant alternative baseline(s) are defined and diagnostics pass.
G10	Closure / Mechanism	The claimed global path or feedback mechanism satisfies adapter closure rules.
G11	Robustness	Material result survives preregistered sensitivity/specification tests.
G12	Falsification	Required negative controls, placebos and adversarial tests are completed.
G13	Contradiction	Material contradictory evidence is resolved or explicitly limits the conclusion.
G14	Reproduction	The result reproduces from frozen artifacts in the required environment.
G15	Claim Escalation	The final statement is no stronger than the

		complete proof chain allows.
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Mandatory Gate Chain: downstream promotion is blocked by unresolved mandatory gates



PASS advances | FAIL rejects a route | INCONCLUSIVE preserves uncertainty | BLOCKED stops dependent claims

Figure 4. Gate chain and blocking semantics.

APPENDIX C - Proof Packet Specification

ProofPacket/
claim.json
hypothesis.json
domain.json
scope.json
timeline.json
evidence_manifest.json
contradictory_evidence.json
source_dependency.json
model_spec.json
assumptions.json
counterfactuals.json
causal_graph.json
closure_results.json
gate_records.json
falsification_results.json
sensitivity.json
run_manifest.json
software_manifest.json
review_record.json
hashes.txt
verdict.json
reproduction.md

Each packet MUST be independently reviewable without access to undocumented analyst state. If an external licensed or restricted source cannot be redistributed, the packet stores a cryptographic identity, acquisition metadata and an authorized retrieval reference rather than fabricating a copy.

APPENDIX D - Requirements Traceability Matrix

Requirement	Normative statement	Verification
REQ-SRC-001	Every material source object is versioned and cryptographically identified.	hash/integrity test
REQ-PROV-002	Every derived material output retains a reversible provenance path.	lineage traversal test
REQ-CLAIM-003	Every material conclusion is represented as a ClaimSpec with explicit dependencies.	schema + dependency test

REQ-MODEL-004	Every model output references a versioned ModelSpec and ImplementationSpec.	model/implementation linkage test
REQ-TIME-005	Historical/holdout analyses respect valid-time and knowledge-time cutoffs.	cutoff leakage negative test
REQ-SCOPE-006	Output scope cannot exceed its supporting certificate without a widening certificate.	scope promotion negative test
REQ-GATE-007	Mandatory gate failures block dependent claim promotion.	control-flow property test
REQ-CF-008	Causal claims define a relevant counterfactual or adapter-equivalent contrast.	counterfactual completeness review
REQ-CAUSAL-009	Causal labels require an explicit estimand and identification route.	identification checklist + diagnostics
REQ-FALS-010	High-impact hypotheses have preregistered falsification or adversarial tests.	negative-control suite
REQ-REPRO-011	Material model runs ship a complete Run Manifest and reproduction route.	fresh-environment rerun
REQ-SEC-012	Production access and evidence mutation follow least privilege and immutable audit controls.	security configuration + audit sample
REQ-SRC-013	Every material source object is versioned and cryptographically identified.	hash/integrity test
REQ-PROV-014	Every derived material output retains a reversible provenance path.	lineage traversal test
REQ-CLAIM-015	Every material conclusion is represented as a ClaimSpec with explicit dependencies.	schema + dependency test
REQ-MODEL-016	Every model output references a versioned ModelSpec and ImplementationSpec.	model/implementation linkage test
REQ-TIME-017	Historical/holdout analyses respect valid-time and knowledge-time cutoffs.	cutoff leakage negative test
REQ-SCOPE-018	Output scope cannot exceed its supporting certificate without a widening certificate.	scope promotion negative test
REQ-GATE-019	Mandatory gate failures block dependent claim promotion.	control-flow property test
REQ-CF-020	Causal claims define a relevant counterfactual or adapter-equivalent contrast.	counterfactual completeness review
REQ-CAUSAL-021	Causal labels require an explicit estimand and identification route.	identification checklist + diagnostics
REQ-FALS-022	High-impact hypotheses have preregistered falsification or adversarial tests.	negative-control suite
REQ-REPRO-023	Material model runs ship a complete Run Manifest and reproduction route.	fresh-environment rerun
REQ-SEC-024	Production access and evidence mutation follow least privilege and immutable audit controls.	security configuration + audit sample
REQ-SRC-025	Every material source object is versioned and cryptographically identified.	hash/integrity test
REQ-PROV-026	Every derived material output retains a reversible provenance path.	lineage traversal test
REQ-CLAIM-027	Every material conclusion is represented as a ClaimSpec with explicit dependencies.	schema + dependency test
REQ-MODEL-028	Every model output references a versioned ModelSpec and ImplementationSpec.	model/implementation linkage test
REQ-TIME-029	Historical/holdout analyses respect valid-time and knowledge-time cutoffs.	cutoff leakage negative test
REQ-SCOPE-030	Output scope cannot exceed its supporting certificate without a widening certificate.	scope promotion negative test
REQ-GATE-031	Mandatory gate failures block dependent claim promotion.	control-flow property test
REQ-CF-032	Causal claims define a relevant counterfactual or adapter-equivalent contrast.	counterfactual completeness review
REQ-CAUSAL-033	Causal labels require an explicit estimand and identification route.	identification checklist + diagnostics
REQ-FALS-034	High-impact hypotheses have preregistered falsification or adversarial tests.	negative-control suite
REQ-REPRO-035	Material model runs ship a complete Run Manifest and reproduction route.	fresh-environment rerun
REQ-SEC-036	Production access and evidence mutation follow least privilege and immutable audit controls.	security configuration + audit sample

REQ-SRC-037	Every material source object is versioned and cryptographically identified.	hash/integrity test
REQ-PROV-038	Every derived material output retains a reversible provenance path.	lineage traversal test
REQ-CLAIM-039	Every material conclusion is represented as a ClaimSpec with explicit dependencies.	schema + dependency test
REQ-MODEL-040	Every model output references a versioned ModelSpec and ImplementationSpec.	model/implementation linkage test
REQ-TIME-041	Historical/holdout analyses respect valid-time and knowledge-time cutoffs.	cutoff leakage negative test
REQ-SCOPE-042	Output scope cannot exceed its supporting certificate without a widening certificate.	scope promotion negative test
REQ-GATE-043	Mandatory gate failures block dependent claim promotion.	control-flow property test
REQ-CF-044	Causal claims define a relevant counterfactual or adapter-equivalent contrast.	counterfactual completeness review
REQ-CAUSAL-045	Causal labels require an explicit estimand and identification route.	identification checklist + diagnostics
REQ-FALS-046	High-impact hypotheses have preregistered falsification or adversarial tests.	negative-control suite
REQ-REPRO-047	Material model runs ship a complete Run Manifest and reproduction route.	fresh-environment rerun
REQ-SEC-048	Production access and evidence mutation follow least privilege and immutable audit controls.	security configuration + audit sample
REQ-SRC-049	Every material source object is versioned and cryptographically identified.	hash/integrity test
REQ-PROV-050	Every derived material output retains a reversible provenance path.	lineage traversal test
REQ-CLAIM-051	Every material conclusion is represented as a ClaimSpec with explicit dependencies.	schema + dependency test
REQ-MODEL-052	Every model output references a versioned ModelSpec and ImplementationSpec.	model/implementation linkage test
REQ-TIME-053	Historical/holdout analyses respect valid-time and knowledge-time cutoffs.	cutoff leakage negative test
REQ-SCOPE-054	Output scope cannot exceed its supporting certificate without a widening certificate.	scope promotion negative test
REQ-GATE-055	Mandatory gate failures block dependent claim promotion.	control-flow property test
REQ-CF-056	Causal claims define a relevant counterfactual or adapter-equivalent contrast.	counterfactual completeness review
REQ-CAUSAL-057	Causal labels require an explicit estimand and identification route.	identification checklist + diagnostics
REQ-FALS-058	High-impact hypotheses have preregistered falsification or adversarial tests.	negative-control suite
REQ-REPRO-059	Material model runs ship a complete Run Manifest and reproduction route.	fresh-environment rerun
REQ-SEC-060	Production access and evidence mutation follow least privilege and immutable audit controls.	security configuration + audit sample

APPENDIX E - Failure Modes and Effects Analysis

ID	Failure mode	Severity	Likelihood	Primary controls
F-01	False entity merge	High	Medium	Deterministic identifiers first; uncertainty; reversible resolution; human review.
F-02	Source echo counted as independent evidence	High	High	Dependency graph, primary-source clustering, duplicate fingerprints.
F-03	Post-cutoff leakage	Critical	Medium	Bitemporal store, frozen cutoffs, automated negative tests.
F-04	Model applied outside validity domain	Critical	Medium	Executable validity predicates and scope gate.
F-05	Correlation promoted to causation	Critical	Medium	Identification gate and claim firewall.
F-06	Benefit/access promoted to	Critical	Medium	Typed relation semantics and

	intent/control			prohibited-inference rules.
F-07	Counterfactual chosen after seeing outcome	High	Medium	Protocol freeze and counterfactual versioning.
F-08	Multiple-testing inflation	High	High	Preregistered search universe, FDR/FWER policy, holdout.
F-09	Contradictory evidence omitted	High	Medium	Mandatory contradiction graph and proof-packet field.
F-10	Hidden manual override	Critical	Low	Immutable review events; machine and human states separated.
F-11	Schema drift	High	Medium	Versioned schemas, migration tests, release gates.
F-12	Irreproducible software environment	High	Medium	Container digests, dependency locks, fresh-environment reproduction.
F-13	Uncertainty collapsed into one misleading score	High	Medium	Multidimensional uncertainty model; score prohibition.
F-14	Graph cycle mistaken for causal feedback	High	Medium	ClosureType semantics and edge-level causal gates.
F-15	Unsupported generalization	Critical	Medium	Scope lattice and widening certificate.
F-16	Adversarial source poisoning	High	Low/Medium	Quarantine, source diversity, anomaly checks, signed provenance.
F-17	Parser silently changes extracted meaning	High	Medium	Transform versions, regression corpus, source-to-extraction sampling.
F-18	Backups unusable	High	Low	Scheduled restore tests and immutable release archives.
F-19	Sensitive data exposed in publication	Critical	Low/Medium	Redaction, privacy review, purpose limitation and role-based access.
F-20	Reviewer bias or conflict	High	Medium	Blinded review where applicable; role separation; reviewer metadata.

APPENDIX F - Test Catalogue

Test ID	Suite	Purpose	Expected control behavior
TEST-001	schema	reject malformed or incompatible typed object	FAIL-as-designed
TEST-002	provenance	prove material output traverses to immutable source	BLOCKED-as-designed
TEST-003	scope	reject unsupported widening or universal negative	PASS
TEST-004	temporal	reject post-cutoff knowledge and invalid causal order	FAIL-as-designed
TEST-005	gates	verify blocker propagates to dependent claim	BLOCKED-as-designed
TEST-006	counterfactual	verify baseline/estimand completeness and sensitivity route	PASS
TEST-007	causal	reject causal label when identification assumptions fail	FAIL-as-designed
TEST-008	closure	reject graph cycle lacking validated edge semantics	BLOCKED-as-designed
TEST-009	falsification	verify negative controls can weaken a target hypothesis	PASS
TEST-010	reproduction	reproduce frozen run from manifest in clean environment	FAIL-as-designed
TEST-011	schema	reject malformed or incompatible typed object	BLOCKED-as-designed
TEST-012	provenance	prove material output traverses to immutable source	PASS
TEST-013	scope	reject unsupported widening or universal negative	FAIL-as-designed
TEST-014	temporal	reject post-cutoff knowledge and invalid	BLOCKED-as-designed

		causal order	
TEST-015	gates	verify blocker propagates to dependent claim	PASS
TEST-016	counterfactual	verify baseline/estimand completeness and sensitivity route	FAIL-as-designed
TEST-017	causal	reject causal label when identification assumptions fail	BLOCKED-as-designed
TEST-018	closure	reject graph cycle lacking validated edge semantics	PASS
TEST-019	falsification	verify negative controls can weaken a target hypothesis	FAIL-as-designed
TEST-020	reproduction	reproduce frozen run from manifest in clean environment	BLOCKED-as-designed
TEST-021	schema	reject malformed or incompatible typed object	PASS
TEST-022	provenance	prove material output traverses to immutable source	FAIL-as-designed
TEST-023	scope	reject unsupported widening or universal negative	BLOCKED-as-designed
TEST-024	temporal	reject post-cutoff knowledge and invalid causal order	PASS
TEST-025	gates	verify blocker propagates to dependent claim	FAIL-as-designed
TEST-026	counterfactual	verify baseline/estimand completeness and sensitivity route	BLOCKED-as-designed
TEST-027	causal	reject causal label when identification assumptions fail	PASS
TEST-028	closure	reject graph cycle lacking validated edge semantics	FAIL-as-designed
TEST-029	falsification	verify negative controls can weaken a target hypothesis	BLOCKED-as-designed
TEST-030	reproduction	reproduce frozen run from manifest in clean environment	PASS
TEST-031	schema	reject malformed or incompatible typed object	FAIL-as-designed
TEST-032	provenance	prove material output traverses to immutable source	BLOCKED-as-designed
TEST-033	scope	reject unsupported widening or universal negative	PASS
TEST-034	temporal	reject post-cutoff knowledge and invalid causal order	FAIL-as-designed
TEST-035	gates	verify blocker propagates to dependent claim	BLOCKED-as-designed
TEST-036	counterfactual	verify baseline/estimand completeness and sensitivity route	PASS
TEST-037	causal	reject causal label when identification assumptions fail	FAIL-as-designed
TEST-038	closure	reject graph cycle lacking validated edge semantics	BLOCKED-as-designed
TEST-039	falsification	verify negative controls can weaken a target hypothesis	PASS
TEST-040	reproduction	reproduce frozen run from manifest in clean environment	FAIL-as-designed
TEST-041	schema	reject malformed or incompatible typed object	BLOCKED-as-designed
TEST-042	provenance	prove material output traverses to immutable source	PASS
TEST-043	scope	reject unsupported widening or universal negative	FAIL-as-designed
TEST-044	temporal	reject post-cutoff knowledge and invalid causal order	BLOCKED-as-designed
TEST-045	gates	verify blocker propagates to dependent claim	PASS
TEST-046	counterfactual	verify baseline/estimand completeness and sensitivity route	FAIL-as-designed
TEST-047	causal	reject causal label when identification assumptions fail	BLOCKED-as-designed
TEST-048	closure	reject graph cycle lacking validated edge semantics	PASS
TEST-049	falsification	verify negative controls can weaken a	FAIL-as-designed

		target hypothesis	
TEST-050	reproduction	reproduce frozen run from manifest in clean environment	BLOCKED-as-designed
TEST-051	schema	reject malformed or incompatible typed object	PASS
TEST-052	provenance	prove material output traverses to immutable source	FAIL-as-designed
TEST-053	scope	reject unsupported widening or universal negative	BLOCKED-as-designed
TEST-054	temporal	reject post-cutoff knowledge and invalid causal order	PASS
TEST-055	gates	verify blocker propagates to dependent claim	FAIL-as-designed
TEST-056	counterfactual	verify baseline/estimand completeness and sensitivity route	BLOCKED-as-designed
TEST-057	causal	reject causal label when identification assumptions fail	PASS
TEST-058	closure	reject graph cycle lacking validated edge semantics	FAIL-as-designed
TEST-059	falsification	verify negative controls can weaken a target hypothesis	BLOCKED-as-designed
TEST-060	reproduction	reproduce frozen run from manifest in clean environment	PASS
TEST-061	schema	reject malformed or incompatible typed object	FAIL-as-designed
TEST-062	provenance	prove material output traverses to immutable source	BLOCKED-as-designed
TEST-063	scope	reject unsupported widening or universal negative	PASS
TEST-064	temporal	reject post-cutoff knowledge and invalid causal order	FAIL-as-designed
TEST-065	gates	verify blocker propagates to dependent claim	BLOCKED-as-designed
TEST-066	counterfactual	verify baseline/estimand completeness and sensitivity route	PASS
TEST-067	causal	reject causal label when identification assumptions fail	FAIL-as-designed
TEST-068	closure	reject graph cycle lacking validated edge semantics	BLOCKED-as-designed
TEST-069	falsification	verify negative controls can weaken a target hypothesis	PASS
TEST-070	reproduction	reproduce frozen run from manifest in clean environment	FAIL-as-designed
TEST-071	schema	reject malformed or incompatible typed object	BLOCKED-as-designed
TEST-072	provenance	prove material output traverses to immutable source	PASS
TEST-073	scope	reject unsupported widening or universal negative	FAIL-as-designed
TEST-074	temporal	reject post-cutoff knowledge and invalid causal order	BLOCKED-as-designed
TEST-075	gates	verify blocker propagates to dependent claim	PASS
TEST-076	counterfactual	verify baseline/estimand completeness and sensitivity route	FAIL-as-designed
TEST-077	causal	reject causal label when identification assumptions fail	BLOCKED-as-designed
TEST-078	closure	reject graph cycle lacking validated edge semantics	PASS
TEST-079	falsification	verify negative controls can weaken a target hypothesis	FAIL-as-designed
TEST-080	reproduction	reproduce frozen run from manifest in clean environment	BLOCKED-as-designed

APPENDIX G - Release Checklist

182. Protocol, hypotheses, data cutoff and thresholds are frozen and signed.
183. All source objects are hashed, immutable and legally/ethically admissible for the intended use.
184. Provenance coverage for material outputs is 100% or release is blocked.

- 185. Entity resolution and normalization QA are complete.
- 186. Evidence, Claim, Model and Temporal graph releases are versioned.
- 187. Scope and validity gates pass for every material conclusion.
- 188. Counterfactual and causal-identification requirements pass where causal language is used.
- 189. Closure claims satisfy adapter semantics and edge-level evidence requirements.
- 190. Negative controls, placebos, alternative explanations and contradictory evidence have been reviewed.
- 191. Multiplicity policy is applied where relevant.
- 192. Uncertainty and sensitivity results are attached.
- 193. Fresh-environment reproduction is complete.
- 194. Proof Packets are generated and their hashes verified.
- 195. Human review requirements are satisfied; no hidden override exists.
- 196. Security, privacy and publication review pass.
- 197. Signed release manifest is created after final artifacts are frozen.
- 198. Prohibited stronger claims are listed explicitly in the release output.

APPENDIX H - Glossary

Term	Definition
Adapter	A domain-specific semantic and validation package that extends FCEA without weakening core controls.
Claim	A proposition whose admissibility, scope and evidence requirements are explicitly represented.
Closure	A domain-specific property that a path or process returns to an equivalent endpoint under declared semantics.
Counterfactual	A declared alternative condition or world used to define a causal or mechanistic contrast.
Evidence item	A source-linked observation or derived fact; not automatically a conclusion.
Gate	A mandatory control that permits, blocks or limits downstream analytical promotion.
Knowledge time	The time at which the analytical system could legitimately know a fact.
Proof Packet	A self-contained package permitting independent review and reproduction of a material verdict.
Scope	The typed domain, time, population/system, regime, assumptions, model family and precision to which a claim applies.
Valid time	The time interval during which the underlying real-world or model state is asserted to hold.
Widening certificate	A named justification that permits a claim to be generalized to a broader scope.